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Palaeomagnetic properties of a Precambrian metamorphic terrane: the Lewisian complex of the Outer Hebrides, NW Scotland

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ABSTRACT

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The palaeomagnetism of a large area of Precambrian metamorphic basement comprising the Outer Hebrides archipelago in NW Scotland is described. A stable magnetite-held imprint characterises much of the complex. Dispersed easterly steep ('A1') and NE intermediate ('C') components are present in the northern sector but are subordinate to NW positive to NW negative ('A2' → 'A5', 'B') components. NW shallow ('A4') components predominate in the south of the outcrop area. 'A1', 'A2' and 'C' components are confined largely to areas which have escaped tectonism since Scourian (> 2650 Ma) times and are probably a relict record. The 'A4' components are most prominent in areas profoundly affected by the Laxfordian episode and record the peak of a later magnetising event. By correlation with APW from the Laurentian and Fennoscandian Shields, relict components are interpreted as a possible acquisition at 1820–1750 Ma close to the peak of Laxfordian tectonism. The younger components are linked to retrogressive amphibolite facies metamorphism and rapid crustal uplift at 1700–1630 Ma. Anisotropy of susceptibility is variable in magnitude and predominantly foliated. It appears to be controlled mainly by Laxfordian strain although the effect of the earlier Scourian episode is also recorded. A younger lamprophyre dyke suite yields a palaeomagnetic pole from nine sites (349° E, 47° S) dated as mid-Permian by reference to the Eurasian APW path.

The spatial distribution of magnetisations in the Outer Hebrides basement mirrors the distribution observed in the Mainland outcrop comprising the NW foreland to the Caledonian orogen but is displaced laterally. The two distributions can be matched by adjustment for ca. 95 km of sinistral strike slip on the site of the Minches Basin between the two outcrops. This fault zone is the NW member of a suite of NE–SW trending transcurrent faults active in late Caledonian times and possesses the largest strike-slip offset.

Introduction

Metamorphic basement complexes possess magnetic remanence acquired during slow uplift-related cooling below the Curie point. This remanence postdates the last ductile phases of deformation and is a sensitive indicator of late events in the consolidation of these complexes such as uprising fluids during retrogressive metamorphism. The Lewisian terrane of NW Scotland comprises a Central Zone preserving structures referred to the Badcallian episode (2800–2600 Ma) of relict granulite facies. It is bordered by broad zones to the north and south (the Northern and Southern Zones) comprehensively reworked during the Laxfordian episode (ca. 1900–1700 Ma) and retrogressed to amphibolite-facies. The Lewisian basement of the Outer Hebrides (Fig. 1) is separated from this outcrop by the Minches, an extensional basin of Mesozoic age, and the relationship of the Lewisian in the Outer Hebrides to the mainland outcrop is a matter of dispute (Dearnley, 1962; Coward and Park, 1987).

The Mainland terrane preserves relict remanence probably acquired near the peak of the Laxfordian episode (Smith and Piper, 1982; Pop-

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pleton and Piper, 1989). More commonly, however, this is partially or completely overprinted by a northwesterly-directed remanence imparted during later event(s) (Poppleton and Piper, 1989; Piper, 1992). The incidence of overprinting varies markedly from place to place both within sampling sites with dimensions of the order of a few metres, and on an areal scale of the order of tens of kilometres. In general it is more prevalent in zones which experienced profound tectonic reworking during the Laxfordian episode; it becomes ubiquitous towards the south in a region which was also the ancient margin of the Laurentian Shield (Bridgwater et al., 1973; Piper, 1990).

Palaeomagnetic investigations of the Lewisian have hitherto concentrated on the Mainland outcrop of NW Scotland (Piper, 1979; Smith and Piper, 1982; Poppleton and Piper, 1989) where

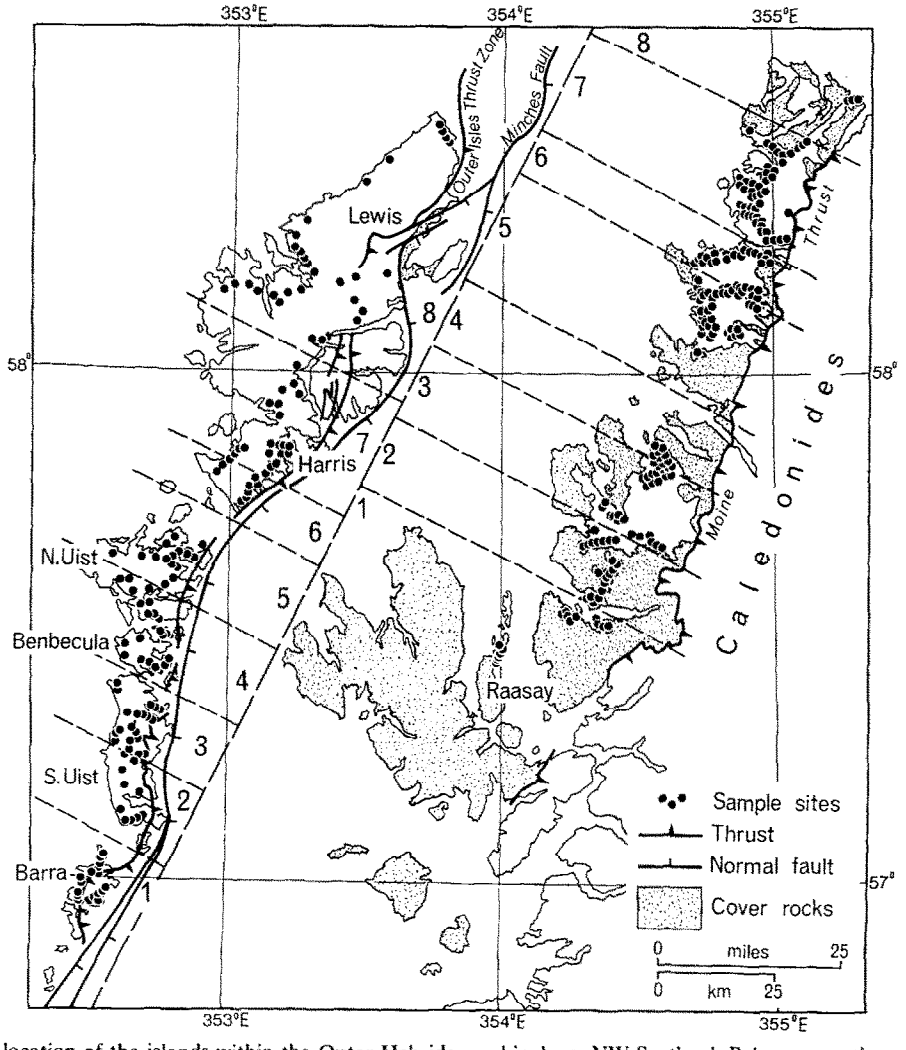


Fig. 1. Regional location of the islands within the Outer Hebrides archipelago, NW Scotland. Palaeomagnetic sampling sites here and on the island of Raasay are the subject of this study. Sampling sites of earlier investigations of the Lewisian (Precambrian) outcrop on the mainland (Piper, 1979; Smith and Piper, 1982; Poppleton and Piper, 1989) are also shown. Note that this outcrop comprises the foreland to the Caledonian orogen in NW Scotland. The two fault zones shown in the Outer Hebrides (after Brewer and Smythe, 1986) are the Outer Isles Thrust Zone, a complex zone of thrust and normal movements with a history commencing in mid-Proterozoic times (Fettes and Mendum, 1987), and the Minches Fault, a major normal fault bordering a zone of Mesozoic extension within the Minches Basin between the Outer Hebrides and the Mainland. The eight numbered zones in the Outer Hebrides and Mainland outcrops of the Lewisian basement are used for regional assessment of magnetic properties. For regional location see Fig. 21.

the Precambrian platform comprises the NW foreland of the Caledonian orogenic belt (ca. 550–400 Ma). This study reports a regional investigation of the Lewisian outcrop in the Outer Hebrides. It has been undertaken to evaluate the magnetic imprint in the basement here, to relate this imprint to the Precambrian metamorphic history, and to relate this terrane to the outcrop on the mainland. The results have implications to both the Precambrian and the Palaeozoic (late Caledonian) tectonic history of this region.

Geological outline

The Outer Hebrides archipelago is the largest area of Lewisian rocks exposed in the British Isles (Fig. 1). It comprises polymetamorphic terrane falling into four major lithological divisions: (a) a basement complex of quartzo–feldspathic gneisses incorporating a range of metasedimentary and metabasic rocks; (b) basic and ultrabasic intrusive rocks belonging to the Scourie dyke suite and comprising about 4 percent of the volume of the complex (Watson and Lisle, 1973); (c) the South Harris Igneous Complex emplaced early in the Laxfordian episode (Dearnley, 1962; Cliff et al., 1983); and (d) granite complexes emplaced late in this episode (Fettes and Mendum, 1987).

Division (a) is predominantly of igneous (tonalite, granodiorite and anorthosite) parentage and derived at ca. 2900–2800 Ma from an upper mantle source. Subsequent deformation resulted in gneiss formation. Following emplacement of a suite of diorite intrusives, the complex was deformed into NNE–SSW folds in late Scourian (2600–2400 Ma) times. Division (b) comprises peridotite, picrite, norite and (most abundantly) microdolerite dykes, sheets and lenses. The earliest members were emplaced into active late Scourian NW–SE shears, while later members are preserved as discordant dykes. Petrographic analogies and age data (Lambert et al., 1970a) link these intrusives with the Scourie dyke suite on the Mainland. They were emplaced into hot crust [ca. 700 °C and 6–7 kbar equivalent to ca. 25 km according to Fettes and Mendum (1987)]. Igneous minerals quickly began recrystallising into metamorphic assemblages although interiors of some thick dykes have escaped recrystallisation (O'Hara, 1961; Dearnley, 1973). The metamorphic condition of the intrusives (equivalent to upper amphibolite grade) is similar throughout the Outer Hebrides and implies that the complex had not undergone appreciable uplift since formation some 500 Ma earlier (Fettes and Mendum, 1987).

Little uplift took place before Laxfordian times because the South Harris Igneous Complex of division (c) of early Laxfordian age (2060–1870 Ma, Cliff et al., 1983) records high-pressure granulite facies metamorphism. This body comprises a layered igneous sequence of anorthosite, gabbros, tonalites and basic dykes deformed during emplacement (Dearnley, 1973). The submarine continuation of the complex to the west is suggestive of disruption by a WNW–ESE trending Laxfordian shear (Westbrook, 1974).

The Laxfordian episode embraces a sequence of structural and metamorphic events including four phases of ductile deformation postdating emplacement of divisions (b) and (c). Scourian fabrics are rotated into parallelism with high-strain zones of folding and shear zone development mostly oriented NW–SE; the Scourie dyke members are correspondingly flattened and disrupted. Although evident throughout the island chain, the magnitude of bulk Laxfordian strain is highly variable even within individual outcrops (Coward, 1973) and is concentrated in the (hydrous) amphibolite zones. Dearnley (1962) considered that North Uist, Benbecula and the northern part of South Uist were devoid of Laxfordian structures in a manner analogous to the Central Zone on the Mainland. Graham and Coward (1973) recognise a more complex pattern of deformation. Whilst zones of anomalously low Laxfordian strain occur in these areas, only small dispersed kernels throughout the island chain have resisted Laxfordian reworking. The Laxfordian episode closed with widespread emplacement of granites of division (d) present as vein and sheet complexes reaching their greatest development in north Harris and south Lewis. This event heralded a phase of rapid uplift and cooling between ca. 1700 and 1500 Ma, recorded by Rb–

Sr mineral ages (Giletti et al., 1961; Lambert et al., 1970a) and K–Ar ages (Lambert et al., 1970b).

The Outer Hebrides Thrust Zone (Figs. 1 to 4) is a complex series of thrusts and crush zones extending along the full length of the eastern seaboard. Fettes and Mendum (1987) link its initiation at depths of 10–15 km to a compressional phase early in the post-Laxfordian uplift. Mylonites, pseudotachylites and cataclastic rocks are present within this zone (Sibson, 1975). Owing to its proximity (ca. 90 km) to the Caledonian orogenic margin, it was reactivated again in Lower Palaeozoic times. Fault products of both post-Laxfordian and Palaeozoic age are identified by K–Ar (Rex and Moorbath in Sibson, 1975) and palaeomagnetic (Piper and Poppleton, 1988) data.

Palaeomagnetic results

Sampling and measurements

Since this terrane was deformed prior to depression of regional temperatures below the Curie point and it is only locally overlain by much younger deposits (Storetvedt and Steel, 1977), the scope for palaeomagnetic field tests is very limited. The best that can be achieved is sampling within the contact aureoles of sparse Permo–Carboniferous and Tertiary dykes which cut the basement. However, palaeomagnetic studies of Lewisian clasts in conglomerates at the base of the Stoer (Piper and Poppleton, 1990) and Torridon (Smith and Piper, 1982; Poppleton and Piper, 1989) Group sedimentary successions on the mainland show that Lewisian rocks are dominated by a remanence predating deposition of these successions at ca. 1100 and 1040 Ma respectively (Smith et al., 1983). Hence there is little doubt that the remanence in the basement of the Outer Hebrides was fixed between the time of the Laxfordian episode (when regional temperatures were in the range 700–500 °C; see Dickinson and Watson, 1976, and O'Hara, 1977) and the time of Torridonian sedimentation.

Although isolated areas of granulite facies are preserved, (Dearnley, 1963), the bulk of the Outer Hebrides outcrop was retrogressed to amphibolite facies during waning stages of the Laxfordian episode. Granulite facies metamorphism takes place above the Curie point and may produce metamorphic magnetite during the prograde phase (Rumble, 1976). Amphibolite facies metamorphism takes place close to, or below, the Curie point and the effects are more complex. A general effect of fluid accession associated with the higher grades of Laxfordian amphibolite facies metamorphism is enhancement of the magnetite content (Piper, 1992). Retrogression to lower grades, as frequently found in shear zones, leads to progressive conversion of magnetite to paramagnetic phases.

Numbers of samples, typically about 7–10, but ranging up to 24, were cored in the field using a portable motorised drill and oriented in situ by sights on the Sun and/or magnetic bearings in conjunction with topographic sights. The bulk comes from members of the Scourie dyke suite and the south Harris igneous complex but many were collected from amphibolite lenses in country rock gneisses; the latter are largely derived from the diorites emplaced in late Scourian times and subsequently deformed and metamorphosed. The total collection comprises 69 sites on Lewis and Harris (Fig. 2), 70 sites on North and South Uist and Benbecula (Fig. 3) and 22 sites on Barra (Fig. 4). The regional relationship of these islands within the island chain can be seen in Fig. 1. Two sites on South Uist come from Tertiary dolerite dykes and contacts, and 9 sites on Barra come from E–W to SE–NW trending quartz dolerite dykes and contacts; these are inferred to belong to a Permo–Carboniferous swarm exposed on the adjacent mainland of Scotland (Jehu and Craig, 1923).

Cores were cut into 2.4 cm long cylinders for laboratory processing. Magnetic anisotropy was determined in 3 or 4 cores from each site. Remanence measurements were made by parastatic or spinner magnetometers and accompanied by stepwise alternating field (a.f.) cleaning incremented in steps of 5 or 10 milliteslas (mT) to peak fields of up to 140 mT. A complementary collection, often cut from the same field cores, was progressively thermally demagnetised in steps of 100 °C to 500 °C and then in 20 °C steps to 600 °C. Although both methods generally proved

satisfactory for resolving the magnetite-held components in these rocks, the thermal data yield the distribution of blocking-temperature components and will comprise most of the descriptive section. Components of magnetisation were recognised by study of vector plots and calculated by principal component analysis (Kirschvink, 1980).

Magnetite is the first phase to seal microfractures within the Lewisian complex (Hay et al., 1988) and is linked to the metamorphic history of the complex; deposition of hematite and paramagnetic minerals occurred during Torridonian and later events. Crack fillings are classed as (a) boundary cracks with lengths up to 1200 μm and width up to 9 μm; (b) intragranular cracks confined within single crystals with lengths up to 4500 μm and widths of up to 25 μm; (c) intragranular cracks crossing boundaries with lengths up to 25 mm and widths up to 1500 μm (Hay et al., 1988). Length-to-width ratios are typically very high and shape anisotropies of the blade-like forms predict high magnetostatic energies. As a result, fairly high proportions of Lewisian metamorphic rocks (ranging from 63 to 97 percent, see Table 2) exhibit magnetic stability (Piper, 1979; Smith and Piper, 1982).

All resolved components are resident in magnetite: this is evident both from the remanence unblocking close to the Curie point of pure magnetite (Figs. 6 and 7) and by the general ability of a.f. cleaning to subtract components (Fig. 8). Present field contributions are not prominent in the NRM record. This is illustrated in Fig. 5 for three zones from the Hebridean outcrop. Only in the northernmost zone (8) is there a large population of steep positive directions similar to the inclination (72°) of the present field. However, most have easterly rather than northerly directions and only small numbers correlate with the predicted Phanerozoic field path. Total NRM components

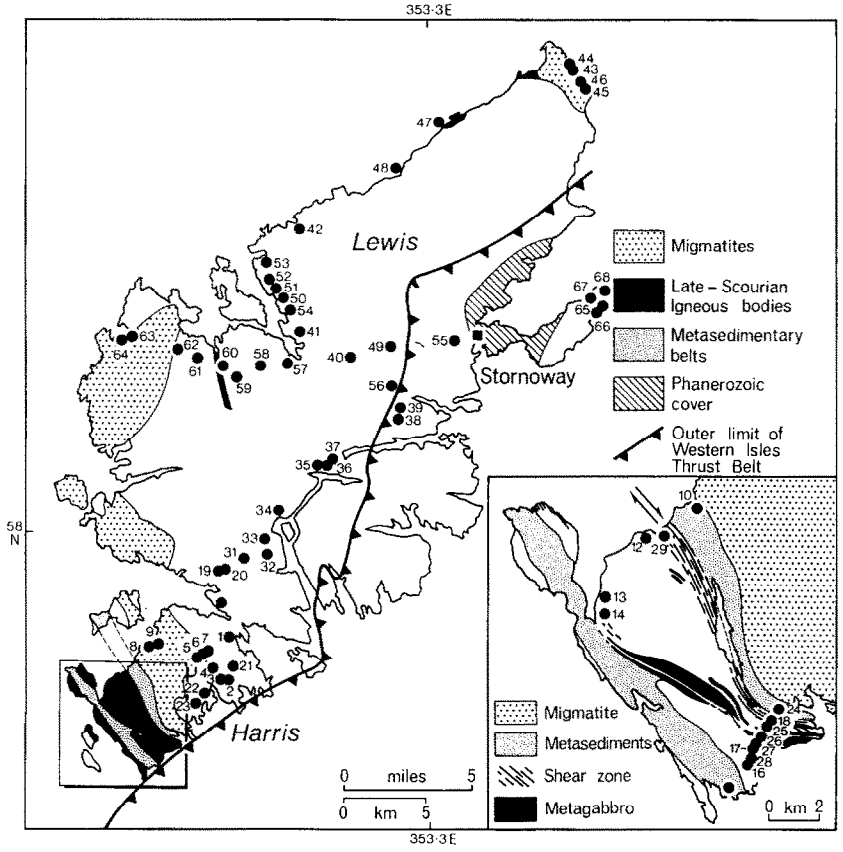


Fig. 2. Palaeomagnetic sampling sites and simplified geology of the islands of Lewis and Harris, Outer Hebrides.

in the southern zone (1) show a preference for the NW-quadrant which also emerges from thermal and a.f. cleaning. Hence remanence carriers do not appear to be susceptible to viscous remanence acquisition. A Palaeozoic or older origin is specifically indicated by a general absence of correlation between the NRM and the palaeofield migration path since 300 Ma predicted from the APW (apparent polar wander) path for Eurasia (Fig. 5).

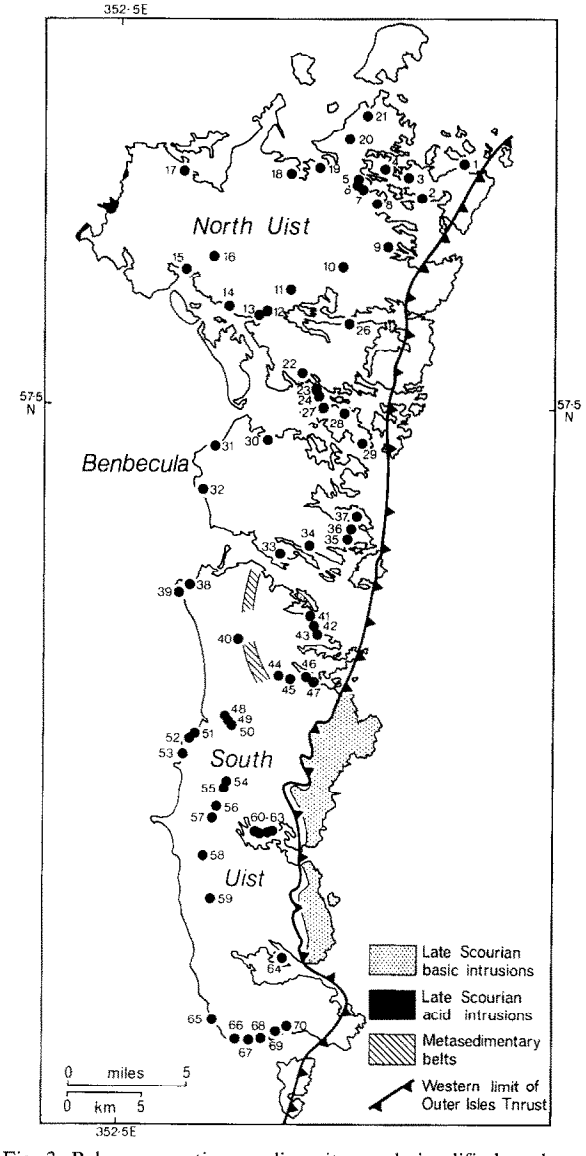


Fig. 3. Palaeomagnetic sampling sites and simplified geology of the islands of North and South Uist and Benbecula, Outer Hebrides.

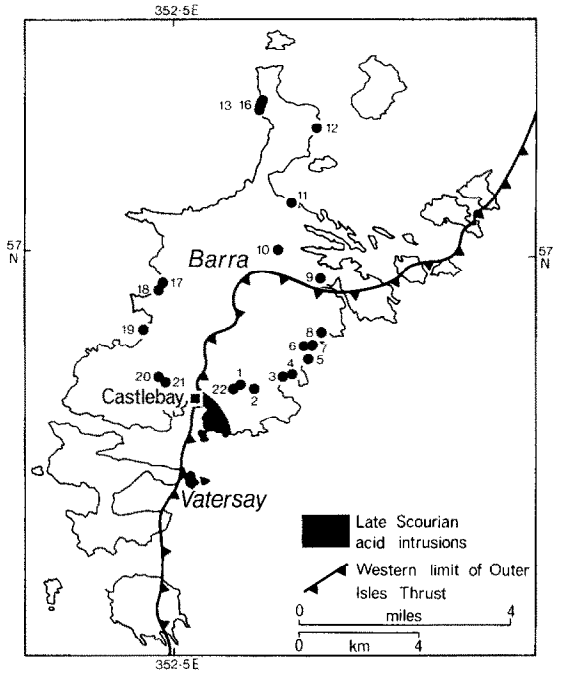


Fig. 4. Palaeomagnetic sampling sites on the Island of Barra, Outer Hebrides. For regional location see Fig. 1.

Since metamorphic terranes cool slowly over protracted periods of time, the composition of the remanence is complex both within sites and between sites. Magnetic components resolved in the Outer Hebrides are (as in the Mainland Lewisian) predominantly directed towards easterly positive (E +) or northwesterly positive to negative (NW + to NW −) directions. They are classified according to their location within 40 × 40° splines of the stereonet according to earlier classifications (Piper, 1979; Smith and Piper, 1982; Fig. 9). Components with easterly steep directions are classed as 'A1' ($D = 50°$ to 130° E, $I = 50°$ to 90°) and with north to northeasterly (N–NE) and intermediate to shallow directions are 'C' ($D = 10°$ to 50° E, $I = 15°$ to 55°); the latter appear to be of one polarity only. Northwesterly directed magnetisations record a continuum of magnetisation commencing with 'A2' ($D = 270°$ to 310° E, $I = 30°$ to 70°) first recognised in relict Scourian terranes on the Mainland (Beckmann, 1976; Piper, 1979). This passes via 'A3' ($D = 305°$ to 345° E, $I = 20°$ to 60°), 'A4' ($D = 295°$ to 335° E, $I = +20°$ to $-20°$), and 'A5' ($D = 295°$ to 335° E, $I = -10°$ to $-50°$)

to 'B' ($D = 275°$ to $315°$ E, $I = -10°$ to $-50°$). This scheme embraces 68% of the resolved components. Although they are predominantly of one polarity, antiparallel (reversed) equivalents of each are recognised. Examples of thermal demagnetisation of 'A1' and 'C' components are shown in Fig. 6 (samples 4–8, 29–6 and 32–1) for samples from Lewis and Harris and in Fig. 7 (samples 7–5 and 67–6) in samples from the Uists and Benbecula. Blocking-temperature spectra are mostly distributed with components subtracted over the whole range of treatment, and coercivities are low. Where these components occur combined with another component, they have the lowest blocking temperatures (e.g. 29–6 and 67–6).

In the NW-directed populations, the steeper positive directions belonging to the 'A2' and 'A3'

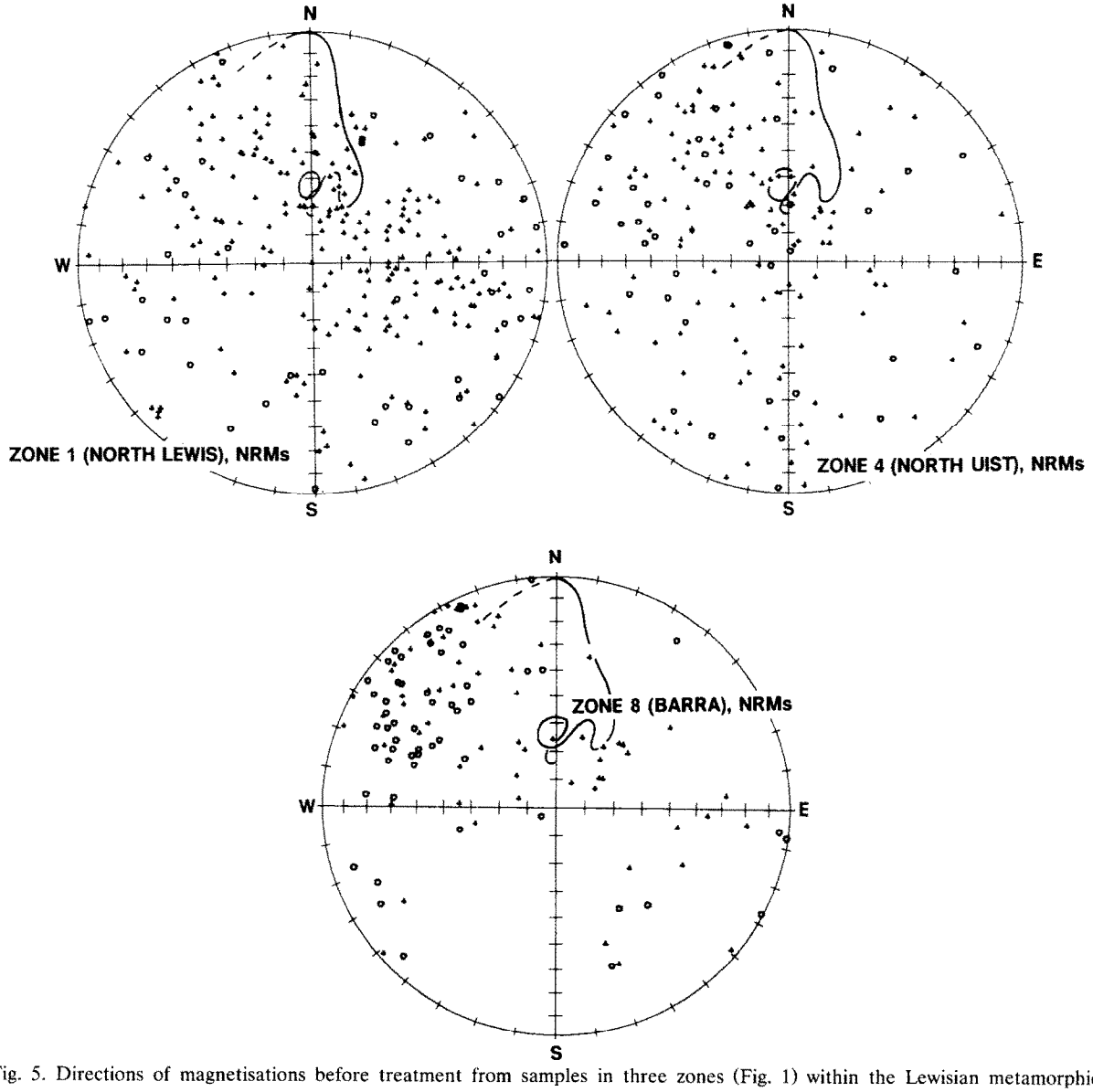


Fig. 5. Directions of magnetisations before treatment from samples in three zones (Fig. 1) within the Lewisian metamorphic basement, Outer Hebrides. Crosses are lower-hemisphere and circles are upper-hemisphere plots. The line (continuous = lower hemisphere; dashed = upper hemisphere) is the predicted path of the palaeofield vector in this area since 310 Ma and is derived from the mean APW path for Eurasia (Piper, 1987b). Equal area projection.

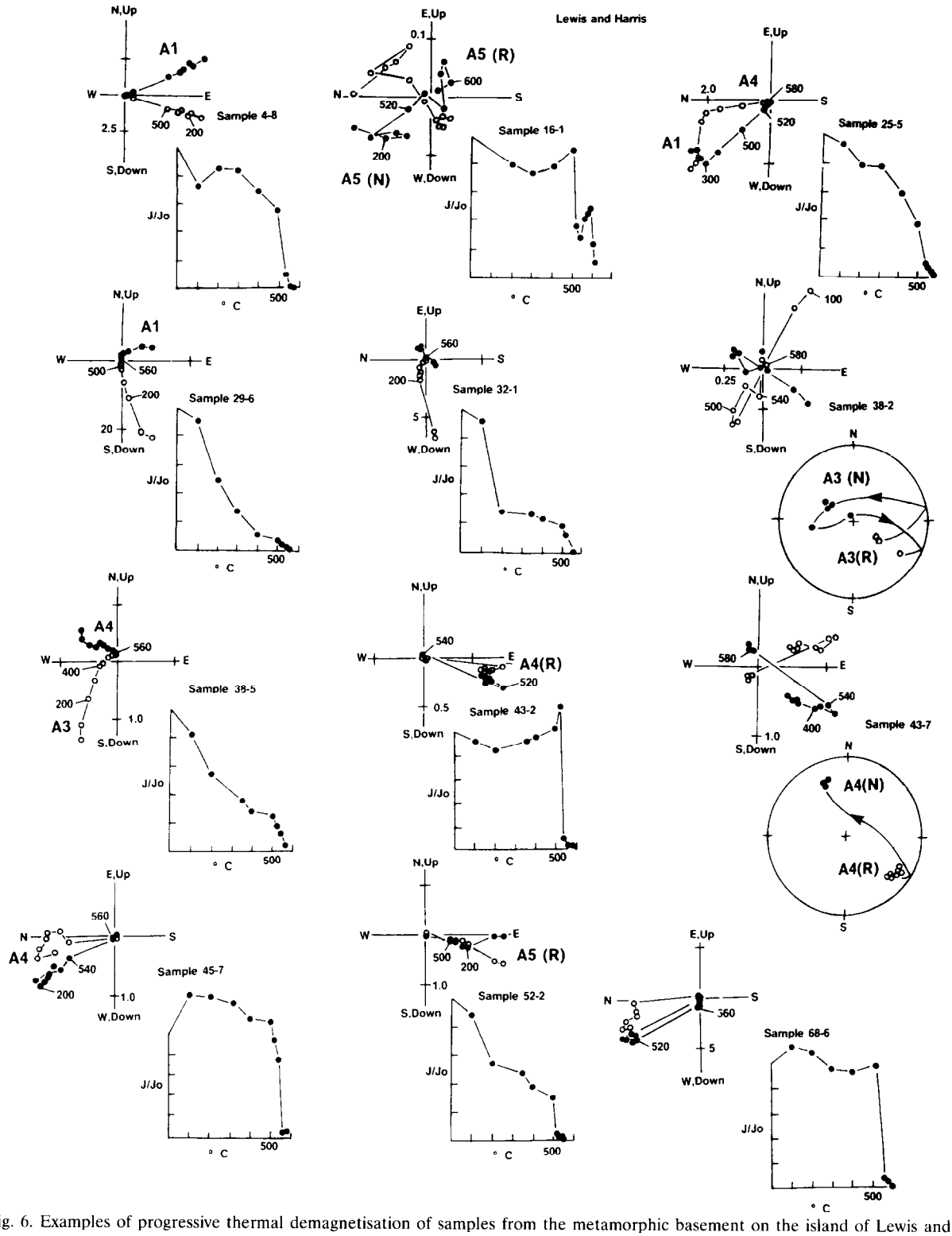


Fig. 6. Examples of progressive thermal demagnetisation of samples from the metamorphic basement on the island of Lewis and Harris, Outer Hebrides, NW Scotland. Closed symbols on the orthogonal plots are projections onto the horizontal plane and open symbols are projection on to the vertical plane. The graphs illustrate decline of the remanence with progressive cleaning as a fraction of the initial NRM (J_o). In the equal area stereonets, the closed symbols refer to lower-hemisphere plots and the open symbols to upper-hemisphere plots. Intensity scales are $\times 10^{-5}$ Am^2 kg^{-1}

populations generally occur as single components but have distributed blocking-temperature spectra (1–8, 17–4, 32–4 and 61–8 in Fig. 7). Sample 38–2 in Fig. 6 is a reversed example of 'A3'. As directions become shallower the blocking-temperature spectra become narrower, although the re-

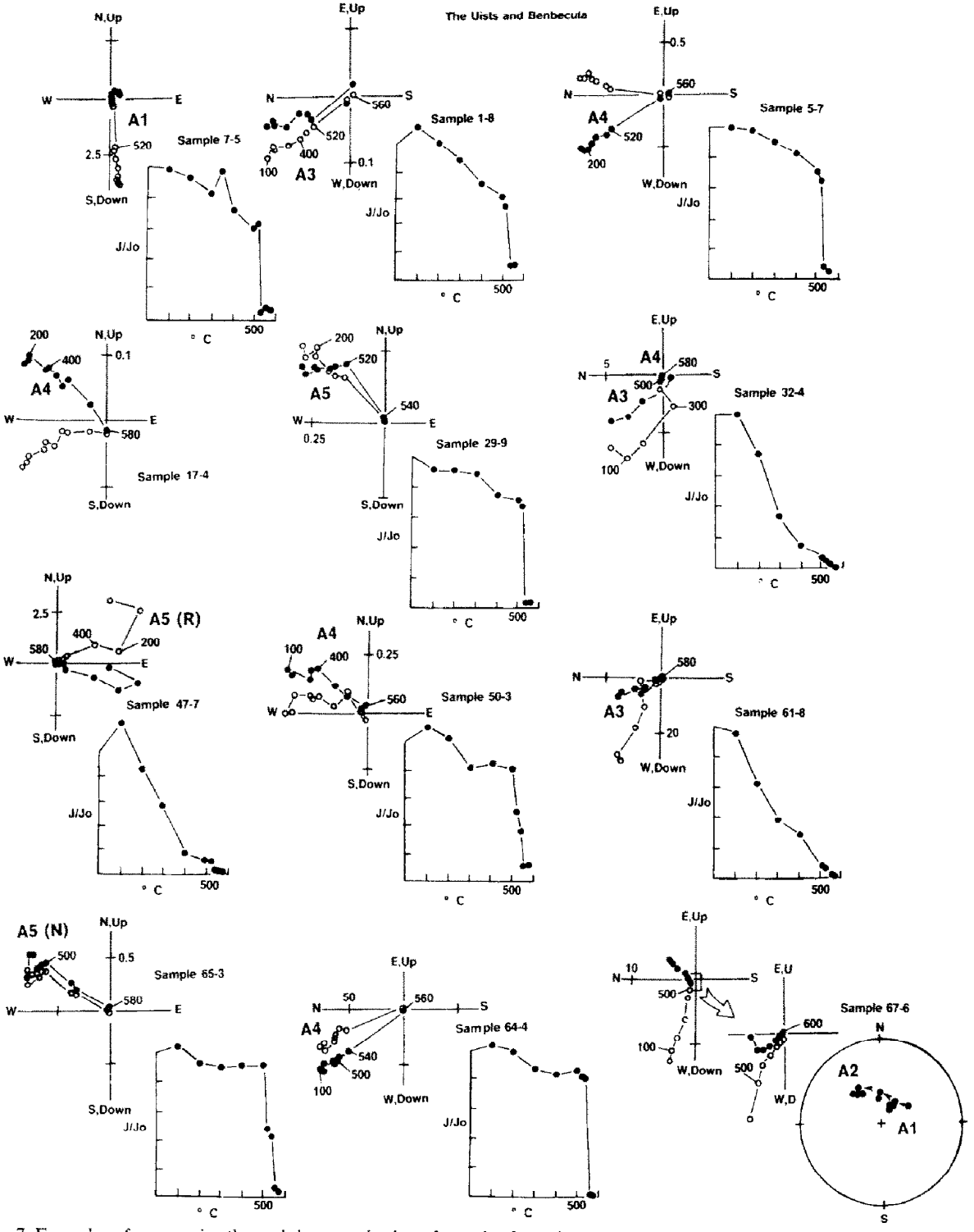


Fig. 7. Examples of progressive thermal demagnetisation of samples from the metamorphic basement on the islands of North and South Uist, Benbecula and Barra in the Outer Hebrides, NW Scotland. The symbols are as for Fig. 6.

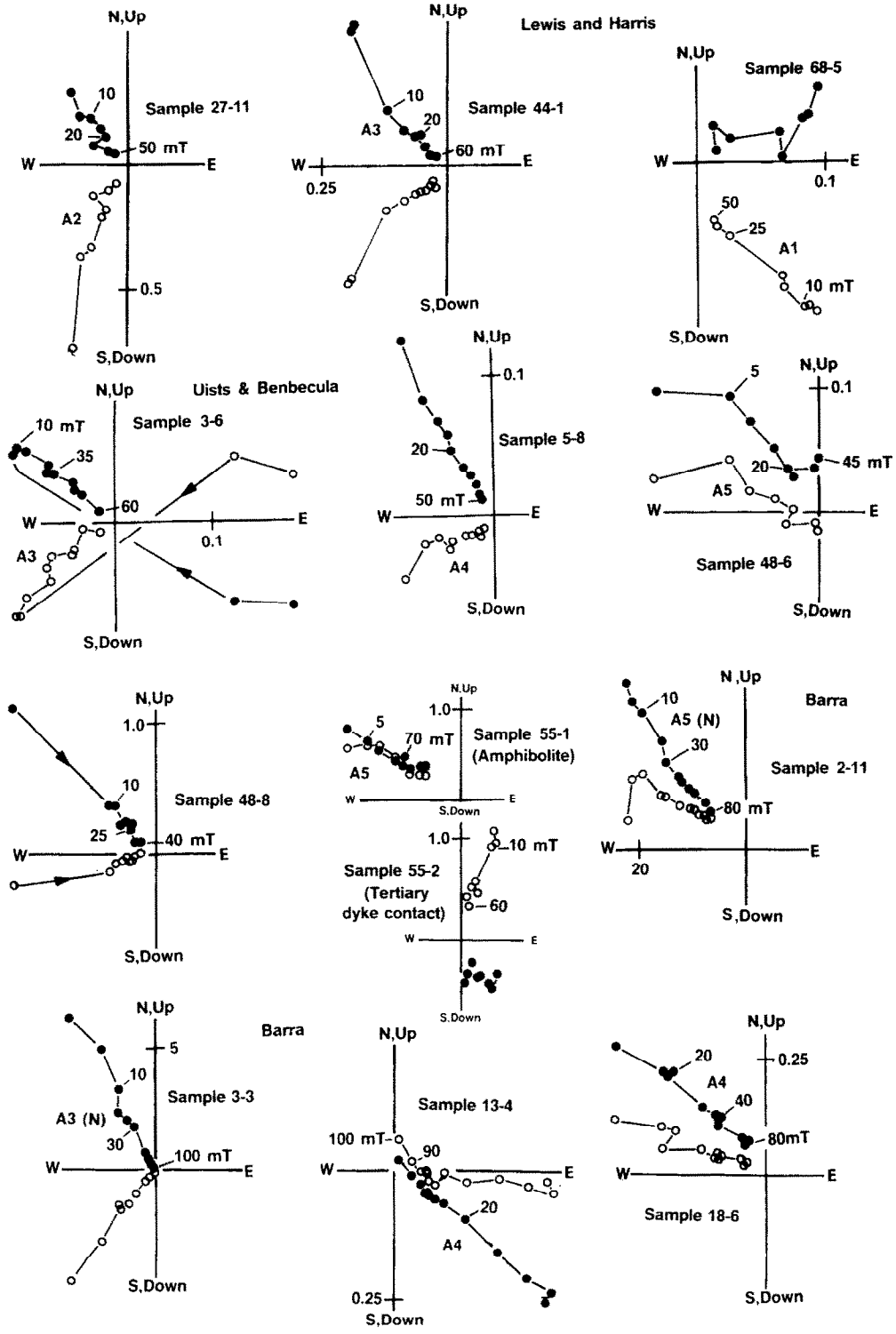


Fig. 8. Examples of alternating field (a.f.) demagnetisation of remanence components in metamorphic rocks from the Outer Hebrides, NW Scotland.

manence continues to be subtracted by the Curie point of magnetite. 'A4' samples with the shallowest inclinations ($I = \pm 10°$) have the most square-shaped J/J_0 versus temperature spectra with the bulk of the remanence subtracted between 540 and 580 °C (samples 45–7 in Fig. 6 and 5–7 and 64–4 in Fig. 7; cf. somewhat steeper inclination examples 17–4 and 50–3 in Fig. 7). Inversions of polarity are most commonly observed in this population either as single components (e.g. sample 43–2 in Fig. 6) or as inversions observed during progressive cleaning of single samples (43–7 in Fig. 6). This implies that magnetisation of individual cores took place over periods much longer than the time required for inversion of the geomagnetic field; they are therefore likely to have integrated short-term variations in the field; hence single cores yield palaeomagnetic (rather than virtual geomagnetic) poles. The 'A4' components are only rarely combined with a second component; in these cases they always reside in the higher blocking-temperature fraction (e.g. samples 25–5 and 38–5 in Fig. 6). Although 'A2' and 'A3' have more distributed blocking-temperature spectra than 'A4', they are sometimes recovered after subtraction of 'A1' (67–6 in Fig. 7) or 'C'; as a general rule 'A1' and 'C' have the lowest blocking temperatures.

As the NW-directed components pass into the negative quadrant, their character becomes more variable. Blocking-temperature spectra of the 'A5' and 'B' components tend to be more distributed (16–1 and 52–2 in Fig. 6 and 47–7 in Fig. 7) but some are comparable to the 'A4' population (29–9 and 65–3 in Fig. 6). Reversed equivalents are also present either as single components (45–2 in Fig. 6 and 47–7 in Fig. 7) or as inversions within the same sample (16–1 in Fig. 6).

Since remanence acquisition is most plausibly linked to uplift-related cooling in Laxfordian or later times, it is anticipated that thermal demagnetisation will be most successful in unravelling the component structure. However, alternating field demagnetisation is also able to define all of the components noted above as discrete entities. Examples illustrated in Fig. 8 include 'A3' components from Lewis and Harris (27–11, 44–1) and a typical low-coercivity 'A1' component (sample 68–5). 'A3', 'A4' and 'A5' components from the Uists include an amphibolite example overprinted by a Tertiary reversed remanence at the contact with a dolerite dyke of this age (cf. 55–1 and 55–2). Both normal and reversed examples of the 'A2' → 'A5' remanence population are recognised (cf. samples 13–4 and 18–6 in Fig. 8).

Examples of cores from the same sample site (38–2 and 38–5 in Fig. 6, and 48–6 and 48–8 in Fig. 8) illustrate typical variability of remanence over short distances. This is a characteristic of acquisition during protracted uplift-related cooling: appreciable polar movement is to be anticipated during the interval of magnetisation and the interpretive strategy must be adapted to accommodate this. In Table 1 all site mean groupings comprising three or more samples yielding a mean with $\alpha_{95} < 25°$ are listed. These means show a distribution (Fig. 9) illustrating the main features of remanence in the Lewisian terrane, but they do not adequately summarise the composition of the remanence within each site. Other components are often present but found only within one or two samples and do not therefore yield an acceptable site mean. This is expressed in terms of within-site coherency where the coherency-parameter is the percentage of the samples from a site defining a single mean direction. This parameter tends to be low in the northern part of the Hebridean outcrop (Table 4) where stable but dispersed directions are found within many sites, and is high in the south where most cores from a site group to yield a common mean direction.

A more representative description of the remanence in the Lewisian basement is given by summing the total number of samples belonging to each population. This is illustrated in histogram-form in Fig. 10. The 'A1' and 'C' groups are not contiguous with the 'A2' to 'A5' and 'B' divisions. They evidently record a separate magnetising event, or events, in this basement. The remaining populations record palaeofield migration during the cooling of the terrane through the range of critical blocking temperatures. This general interpretation is supported by the observation that the number of recorded components increases towards the central part of this contin-

TABLE 1

Site mean palaeomagnetic data from the Lewisian (Precambrian) metamorphic terrane of the Outer Hebrides, NW Scotland

Site no.	D	I	n (N)	R	α_{95}	k
(I) Zone 8 (N. Lewis)						
35	297	39	4 (9)	3.82	23	17
36	92	49	6 (7)	5.65	18	14
38	320	0	5 (9)	4.61	25	10
39	315	34	4 (8)	3.91	17	32
40	324	−9	4 (11)	3.81	24	16
41	103	29	5 (9)	4.83	16	23
43 (i)	116	63	3 (8)	2.96	18	50
43 (ii)	119	−12	7	6.76	12	25
44 (i)	316	42	4 (8)	3.86	20	22
44 (ii)	105	25	3	2.98	13	88
45	327	13	7 (9)	6.58	17	14
46	341	23	6 (8)	5.70	17	17
47 (i)	170	81	4 (10)	3.85	21	20
47 (ii)	104	32	4	3.83	23	18
48	13	54	3 (7)	2.99	10	161
50	334	22	5 (7)	4.83	16	24
53 (i)	45	19	3 (8)	2.94	21	36
53 (ii)	109	28	4	3.97	10	89
54	166	67	3 (7)	2.98	12	113
55 (i)	28	43	3 (8)	2.93	24	28
55 (ii)	299	−3	3	2.94	19	42
56	287	17	5 (7)	4.71	21	14
58	334	51	5 (7)	4.83	16	24
60	283	53	3 (7)	2.94	22	33
62	81	61	5 (8)	4.92	11	52
64 (i)	37	66	3 (7)	2.95	21	37
64 (ii)	309	60	3	2.95	21	37
65	26	38	3 (7)	2.94	23	31
68	38	56	5 (9)	4.86	14	29
(II) Zone 7 (S. Lewis and north N. Harris)						
1 (i)	284	5	7 (24)	6.28	22	8
1 (ii)	314	36	10	9.31	14	13
4 (i)	48	40	6 (14)	5.62	13	13
4 (ii)	333	45	6	5.79	14	24
5	336	4	7 (9)	6.53	18	13
21	285	46	4 (8)	3.79	25	15
23	323	28	6 (8)	5.57	21	12
31	30	28	3 (9)	2.95	21	40
32 (i)	321	50	4 (8)	3.87	20	22
32 (ii)	63	49	5	4.84	16	24
34	47	−54	5 (6)	4.68	23	12
(III) Zone 6 (S. Harris igneous complex)						
11	278	46	9 (9)	8.92	5	101
12	18	−2	5 (9)	4.63	24	11
13	290	57	3 (9)	2.98	12	112
15	312	−20	7 (8)	6.36	21	9
16	312	−36	3 (5)	2.92	25	26
18 (i)	329	19	4 (11)	3.84	21	20
18 (ii)	49	27	3	2.92	25	25
27	304	33	4 (11)	3.82	23	17
28	300	55	5 (9)	4.91	11	46
29	107	68	3 (9)	2.96	17	56

TABLE 1 (continued)

Site no.	D	I	n (N)	R	α_{95}	k
(IV) Zone 5 (N. Uist)						
8	308	26	8 (9)	7.84	8	43
7	61	78	4 (7)	3.94	13	54
6	316	25	5 (7)	4.90	12	39
1 (i)	334	33	8 (9)	7.83	9	41
1 (ii)	2	37	4 (8)	3.95	12	58
2	346	9	7 (7)	6.37	21	10
3	308	−3	8 (8)	7.32	18	10
4	317	−43	11 (12)	10.39	12	16
5	337	7	9 (9)	8.74	9	31
21	334	56	3 (7)	2.97	16	57
19	322	−70	8 (8)	7.62	13	19
17	306	18	3 (9)	2.93	23	29
9	270	54	3 (8)	2.96	17	57
10	329	−25	5 (8)	4.91	12	42
14 (i)	264	−31	3 (8)	2.95	21	34
14 (ii)	274	47	3 (8)	2.94	21	36
(V) Zone 4 (Benbecula and south S. Uist)						
26	332	28	5 (8)	4.92	11	48
32	329	60	4 (9)	3.97	10	87
22	326	0	3 (8)	2.93	23	30
27	2	−20	5 (8)	4.77	19	18
29	326	−30	8 (9)	7.38	17	11
34 (i)	305	35	3 (7)	2.94	22	32
34 (ii)	19	47	3	2.96	17	57
33	157	−45	3 (9)	2.94	21	35
(VI) Zone 3 (north S. Uist)						
48	322	4	6 (9)	5.82	13	27
49 +	211	−74	3 (3)	2.98	13	94
50	307	1	5 (11)	4.76	19	17
54 +	213	−58	3 (3)	2.98	12	107
55	309	−49	7 (9)	6.90	8	62
51 (i)	9	51	4 (8)	3.94	13	52
51 (ii)	318	72	3	2.92	25	25
52	309	50	3 (5)	2.88	31	17
53	328	36	3 (8)	2.94	22	31
44	333	−68	6 (8)	5.87	11	40
46	277	−33	6 (7)	5.74	16	19
47	333	35	5 (9)	4.66	23	12
38	336	33	5 (8)	4.93	10	57
41	304	−43	5 (8)	4.66	23	12
43	299	−39	4 (6)	3.81	24	16
40	307	36	4 (8)	3.81	24	16
+ Tertiary dyke						
(VII) Zone 2 (south S. Uist)						
60	295	−22	8 (9)	7.78	10	32
61 (i)	334	−15	5 (8)	4.69	22	13
61 (ii)	332	36	3	2.92	25	27
62	307	−19	7 (9)	6.63	15	16
63	316	−19	6 (7)	5.64	19	14
56	327	−20	4 (7)	3.91	16	34
64	325	22	7 (9)	6.80	11	31

TABLE 1 (continued)

Site no.	D	I	n (N)	R	α_{95}	k
(VII) Zone 2 (south S. Uist)						
66	317	15	9 (9)	8.73	10	30
67	313	13	5 (8)	4.93	10	60
69	319	−4	6 (7)	5.47	23	9
70	327	−20	7 (7)	6.63	15	16
68	317	−16	5 (8)	4.76	19	16
65	313	−29	4 (8)	3.88	18	26
59	311	−3	6 (8)	5.85	12	33
(VIII) Zone 1 (Barra)						
1A	177	8	7 (9)	6.75	13	24
2	323	−31	9 (10)	8.73	10	30
4A	335	−22	5 (8)	4.75	17	16
5	315	−42	11 (11)	10.48	11	19
6	313	−23	9 (10)	8.51	13	16
9	293	−42	11 (11)	10.50	10	20
10	328	−39	9 (12)	8.51	13	17
11	306	−17	7 (8)	6.69	14	20
15	296	0	4 (9)	3.95	13	55
17	316	−6	8 (10)	7.40	17	12
18	311	−16	11 (12)	10.83	6	58
19	321	0	11 (12)	10.77	7	44
20	292	−17	8 (8)	7.56	14	16
21 +	168	−61	3 (3)	2.93	23	30
22	310	−4	4 (6)	3.84	21	19
+ Tertiary dyke						
(IX) Upper Palaeozoic dykes, Barra						
1B	196	−8	4 (4)	3.85	21	20
3	159	−38	3 (3)	2.93	23	29
4B	187	−42	5 (5)	4.68	22	13
7	177	−15	3 (3)	2.99	5	558
8	189	−25	4 (4)	3.99	5	289
12	169	−16	8 (9)	7.62	13	18
13	191	−36	9 (9)	8.45	14	15
14	184	−22	4 (4)	3.96	11	71
16	201	−30	5 (5)	4.90	12	40
Group mean result, Upper Palaeozoic dykes (352.5 ° E, 57.0 ° N)						
	183.8	−26.3	9 sites	8.67	10.6	24.4
				Palaeomagnetic pole		
				° E	° S	d_p/d_m
				346.1	46.8	6.2/11.5

Note: D and I are the mean declination and inclination derived from n samples from a site totalling N measured samples with resultant vector R and Fisher precision parameter $k = (n-1)/(n-R)$; α_{95} is the cone of 95 percent confidence about the mean direction in degrees; d_p and d_m are the radii of the confidence oval about the pole position in the colatitude direction and at right angles to it, respectively.

uum ('A4') and then declines into 'A5' and 'B'. The character of the remanence changes along this progression with blocking-temperature spectra becoming higher and narrower to 'A4' and then more variable in character beyond this. There is no specific indication of the temporal

TABLE 2

Mean directions of magnetisation and pole positions for magnetisation groups in the Lewisian terrane, Outer Hebrides, NW Scotland

Magnetisation	D	I	N	R	k	α_{95}	Palaeomagnetic pole		
							°E	°N	d_p/d_m
A1 (sites)	95.1	68.4	4	3.9	50.8	13.0	45.9	39.3	18.5/21.9
A1 (samples)	91.5	66.2	44	42.8	34.9	3.7	50.7	38.9	5.0/6.1
A2	292.9	47.8	13	12.7	39.7	6.7	260.8	36.2	5.7/8.7
A3	325.0	38.1	25	24.3	33.1	5.1	222.9	45.7	3.6/6.0
A4	315.3	−3.0	29	28.0	28.2	5.1	221.7	20.9	2.6/5.1
A5	316.6	−28.4	22	21.3	31.5	5.6	215.1	8.9	3.4/6.2
B	298.4	−29.6	17	16.5	30.4	6.6	230.7	0.8	4.0/7.3
C	32.4	40.3	10	9.7	27.5	9.4	125.2	48.2	6.8/11.3
Raasay (N–S)	359.0	2.1	25 *	24.1	27.3	5.7	175.2	33.4	2.8/5.6

Footnote: A1: D = 50–130 ° E, I = +50 −90 °; A2: D = 270–310 ° E, I = +30 −70 °; A3: D = 305–345 ° E, I = +20 −60 °; A4: D = 295–335 ° E, I = +20 to −20 °; A5: D = 295–335 ° E, I = −10 to −50 °; B: D = 275–315 ° E, I = −10 to −50 °; C: D = 10–50 ° E, I = +15 −55 °.

D and I are the mean declination and inclination derived from N sites (* = samples) with resultant vector of magnitude R and Fisher precision parameter $k[=(N-1)/(N-R)]$. α_{95} is the radius of the cone of 95 percent confidence above the mean direction and d_p and d_m are the radii of the semi axes of the cone of confidence about the mean palaeomagnetic pole in the colatitude direction and at right angles to it respectively. (Mean location, Hebrides: 353.0 E, 57.8 N; Raasay: 354.0 E, 57.4 N).

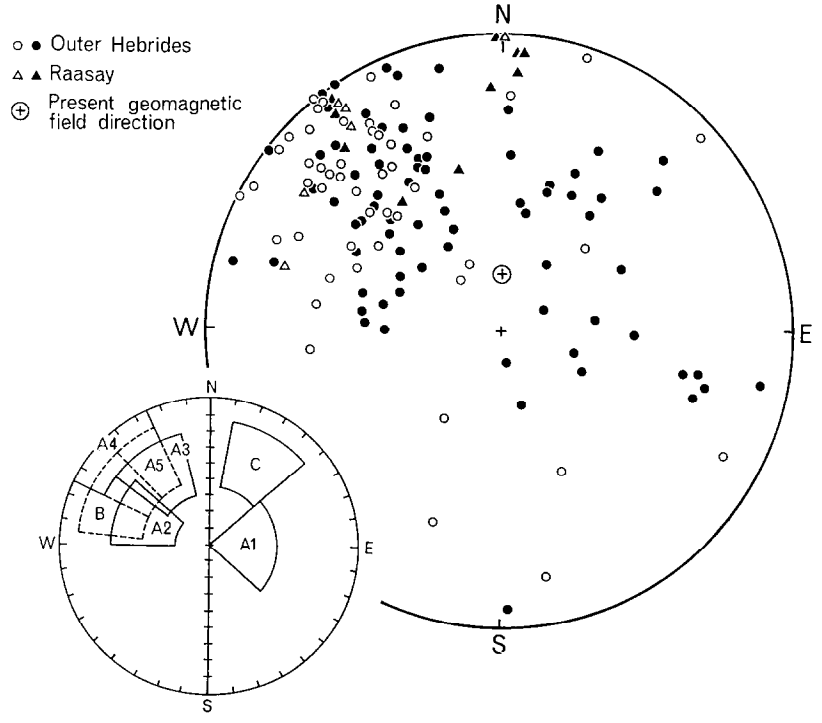


Fig. 9. Site mean directions of magnetisation ($N > 3$, $\alpha_{95} < 25°$) from the Lewisian metamorphic basement of the Outer Hebrides, NW Scotland. Closed symbols are lower-hemisphere plots and open symbols are upper-hemisphere plots. Equal area projection. The inset projection shows the classification of components defined by earlier investigations (Piper, 1979; Smith and Piper, 1982) and employed in this study. All components except 'A1' are referred to 40 × 40 ° splines given in the text.

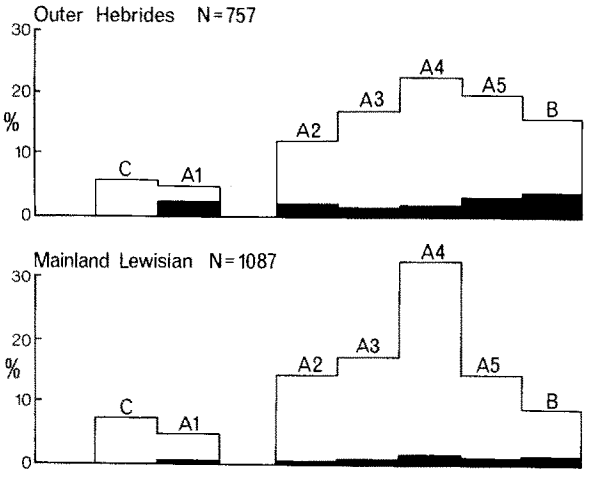


Fig. 10. Histogram showing the percentage populations of samples falling within each of the component subdivisions (Fig. 9) in the Outer Hebrides. The shaded portions are antiparallel (reversed) populations. The percentages of these components in the Mainland Outcrop of the Lewisian basement (from data of: Piper, 1979; Smith and Piper, 1982; Poppleton and Piper, 1989) are shown for comparison. N is the number of samples.

sequence of these magnetisations from the evidence as summarised in Figs. 9 and 10. The blocking-temperature spectra will reflect the character (shape and oxidation state) of the magnetite carriers and are not reliable indicators of relative age. It is argued below from the spatial distribution of these magnetisations that 'A2' is the earliest part of this sequence. The palaeofield migration from 'A2' to 'B' would then imply that this terrane moved from intermediate palaeolatitudes ($\lambda \sim 30°$) towards the palaeoequator, and then moved into the opposite hemisphere ($\lambda \sim 15°$) during the interval recorded by these magnetisations.

The remanence imprint in the metamorphic basement of the Outer Hebrides is comparable to the imprint in the Mainland outcrop (Piper, 1979, 1987a; Smith and Piper, 1982; Fig. 10). The 'C' and 'A1' are subordinate in both areas, while the 'A2' to 'A5' and 'B' populations peak in 'A4'. Small populations of reversed components of all directions except 'C' are also observed (shaded areas in Fig. 10). Hence it may be inferred that both outcrops have experienced essentially the same history of magnetisation.

The island of Raasay in the Inner Hebrides is the southern-most exposed extension of the Lewisian basement north of the Great Glen Fault

TABLE 3

Site mean palaeomagnetic data from the Lewisian Precambrian metamorphic terrane on the Island of Raasay, Inner Hebrides

Site no.	D	I	$n(N)$	R	α_{95}	k
1	319	16	12 (14)	11.73	7	40
2 (i)	305	−16	7 (11)	6.76	12	25
2 (ii)	0	−19	4	3.83	22	18
3 (i)	338	−28	11 (14)	10.50	11	20
3 (ii)	3	7	4	3.82	23	17
4	321	2	9 (11)	8.50	13	16
5	316	−13	6 (6)	5.82	13	28
6	325	−8	6 (8)	5.87	11	38
7	359	2	3 (7)	2.98	11	130
8	322	−13	4 (5)	3.86	20	21
9	322	−3	9 (12)	8.76	9	33
10	6	7	4 (13)	3.93	15	41
11	310	17	3 (7)	2.85	35	13
12 +	211	−76	3 (3)	2.98	11	132
13	4	13	4 (7)	3.83	22	18
14 (i)	356	17	6 (12)	5.85	12	34
14 (ii)	344	40	5	4.74	20	16
17	323	7	5 (11)	4.92	11	49
18	319	−29	7 (12)	6.48	17	12

Symbols are as for Table 1; + = Tertiary sheet.

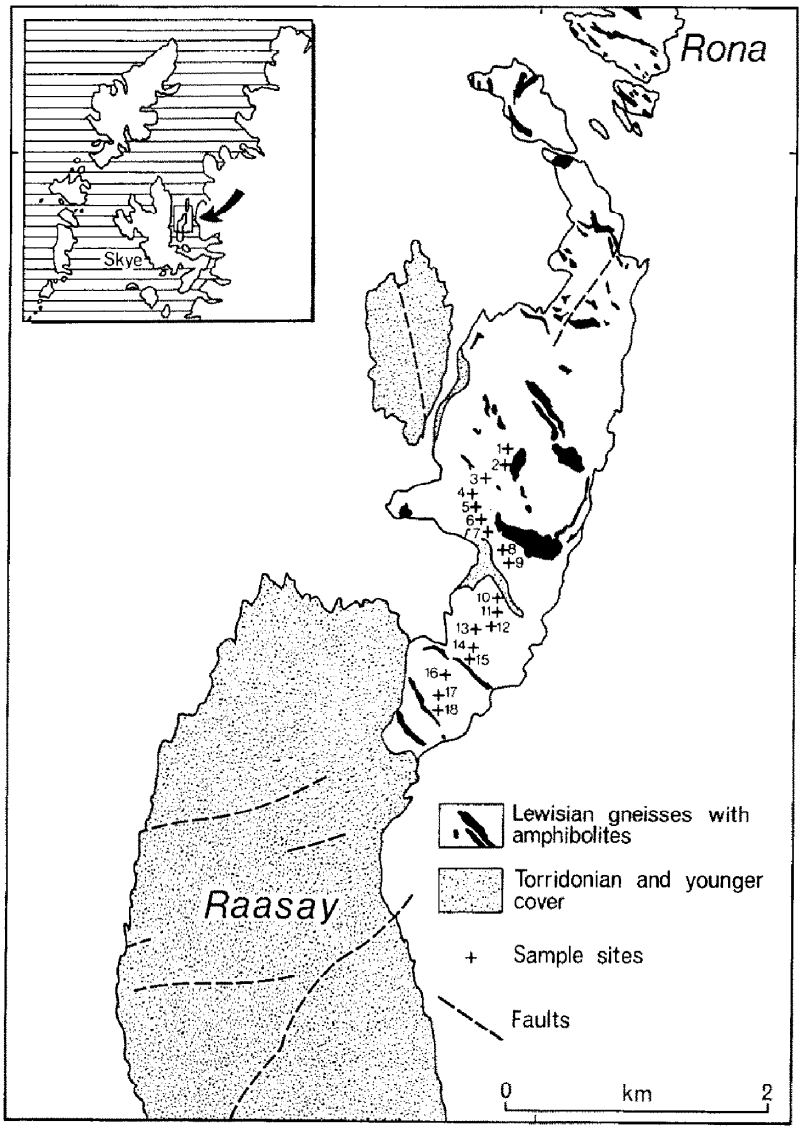


Fig. 11. Palaeomagnetic sampling sites in the Precambrian basement of the island of Raasay, Inner Hebrides, NW Scotland. Note that site numbers indicated as 1 to 18 in this figure refer to sites numbered 40 to 57 in Fig. 12. The regional location of this basement is shown in Fig. 1.

Zone (Fig. 1). It comprises gneisses and Scourie dykes strongly reworked by Laxfordian tectonism and comparable with the Southern Zone of the Mainland Outcrop. For comparison with this zone (cf. Poppleton and Piper, 1989) gneisses and amphibolite bodies have been sampled at 18 locations on Raasay (Fig. 11). Components are also resident in magnetite and most commonly have W to NW directions of shallow inclination. Samples 40–2, 47–10, 56–3 and 57–8 in Fig. 12 are examples. They compare with 'A4' components in the Outer Hebrides and Mainland outcrops which are most widely developed in the south of each area (Fig. 17). Reversals of magnetisation are sometimes present in individual samples (40–2 in Fig. 14).

A feature of the remanence not observed elsewhere is a shallow N–S directed component (46–3, 48–2, 53–6 and 57–8 in Fig. 12) with properties similar to the 'A4' components. Reversals are again found in single cores (44–4). Six of 15 stable sites yield group means with this direction (Table 3); its relative age is uncertain. The presence of a subordinate reversed direction suggests

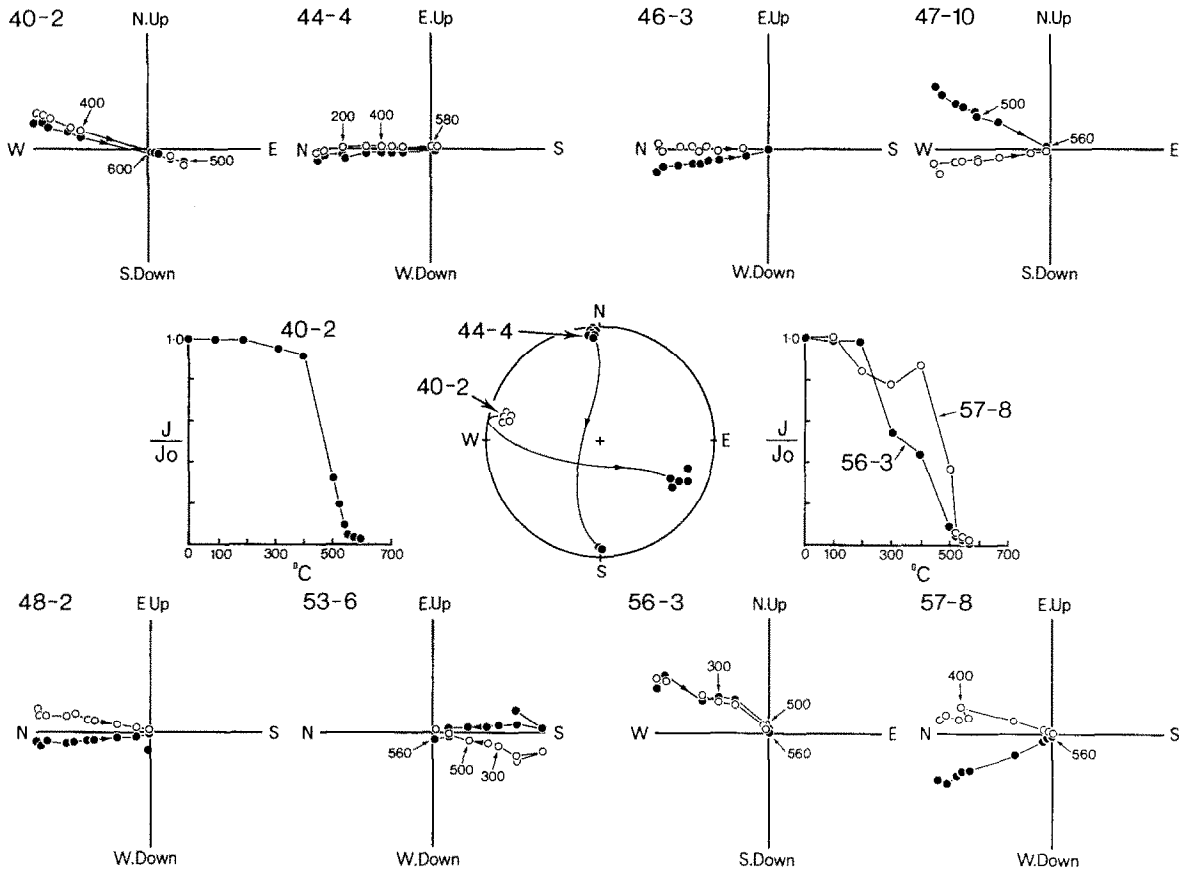


Fig. 12. Examples of thermal demagnetisation results from gneiss and amphibolite samples from the Lewisian terrane on Raasay, Inner Hebrides. The symbols are as for Fig. 6.

protracted acquisition; it may identify a stage in uplift and cooling of the complex not recorded elsewhere.

Magnetic fabric

Measurement and calculation of anisotropy of susceptibility assume that it can be represented by a second-rank symmetrical tensor K_{ij} relating induced magnetisation J_i to the applied field H_j by:

$$J_i = K_{ij} H_j$$

The susceptibility can then be completely specified by six orthogonal measurements yielding maximum (K_{max}), intermediate (K_{int}) and minimum (K_{min}) susceptibility values and corresponding principal axes. In metamorphic rocks, mineral growth takes place preferentially in directions of minimum stress and is expressed as a lineation or foliation. Ferromagnetic and paramagnetic minerals contribute to the bulk susceptibility and provide a measure of this property. A large number of anisotropy parameters has been used to express properties of the ellipsoid (Tarling, 1983) and a need for uniformity in presentation of anisotropy data is recognised (Ellwood et al., 1988). Accordingly, data are represented in terms of the degree of susceptibility anisotropy where:

Degree of anisotropy $= K_1/K_3$

a lineation parameter, L, where:

$$L = K_1/K_2$$

and a foliation parameter, F, where:

$$F = K_2/K_3$$

The latter two parameters are often illustrated as F–L diagrams which bear a close resemblance to Flinn plots.

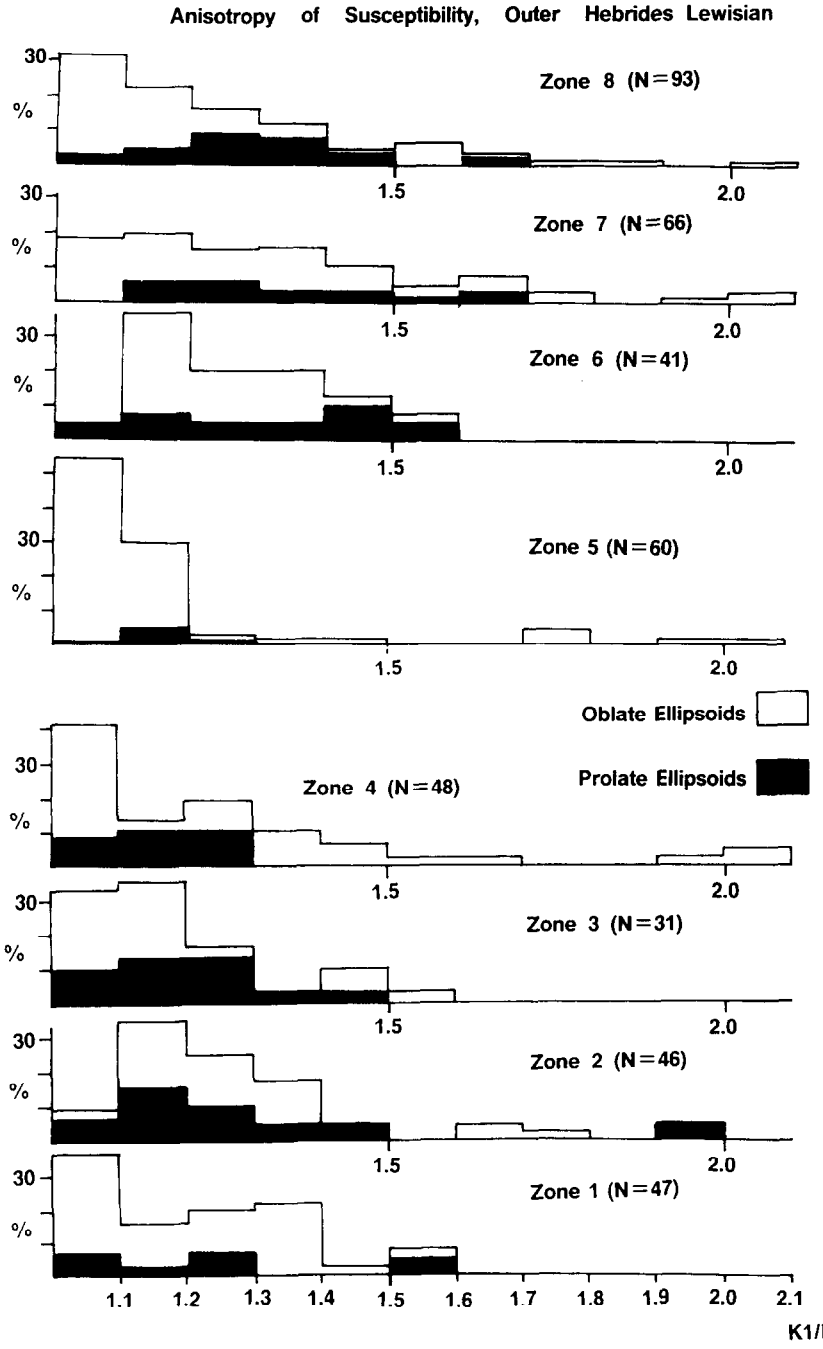


Fig. 13. The magnitude of anisotropy of susceptibility (K_1/K_3) in each of the 8 zones within the Lewisian metamorphic terrane of the Outer Hebrides shown in Fig. 1, expressed in histogram-form as a percentage of the total number of measured cores (N). The susceptibility ellipsoids are distinguished as oblate ($K_2/K_3 > K_1/K_2$) or prolate ($K_1/K_2 > K_2/K_3$).

The degree of anisotropy of susceptibility (K_1/K_3) is illustrated in Fig. 13 for amphibolites and basic gneisses from each of the eight zones in the Hebridean outcrop (Fig. 1). Although magnitudes are predominantly less than 1.3, values as high as 2.1 are locally observed. They are consistently high in zone 6 incorporating the South Harris Igneous Complex (Fig. 2). This is a zone of profound early Laxfordian tectonism (Dearnley, 1962) and incorporates major zones of Laxfordian

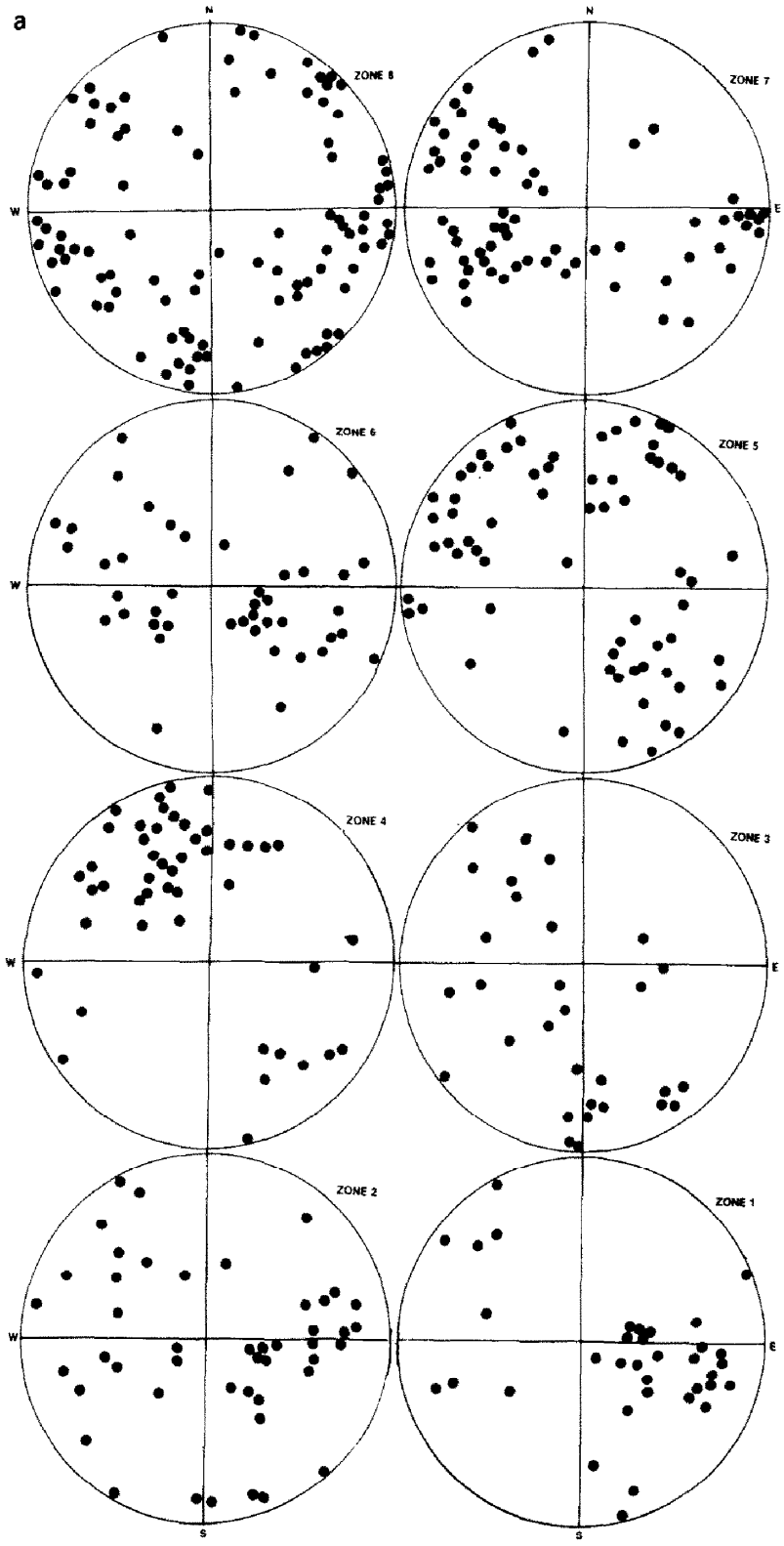


Fig. 14. The poles to (a) foliation of magnetic susceptibility and (b) lineation of magnetic susceptibility within each of the eight zones shown in Fig. 13. The plots are on the lower hemispheres of equal area projections.

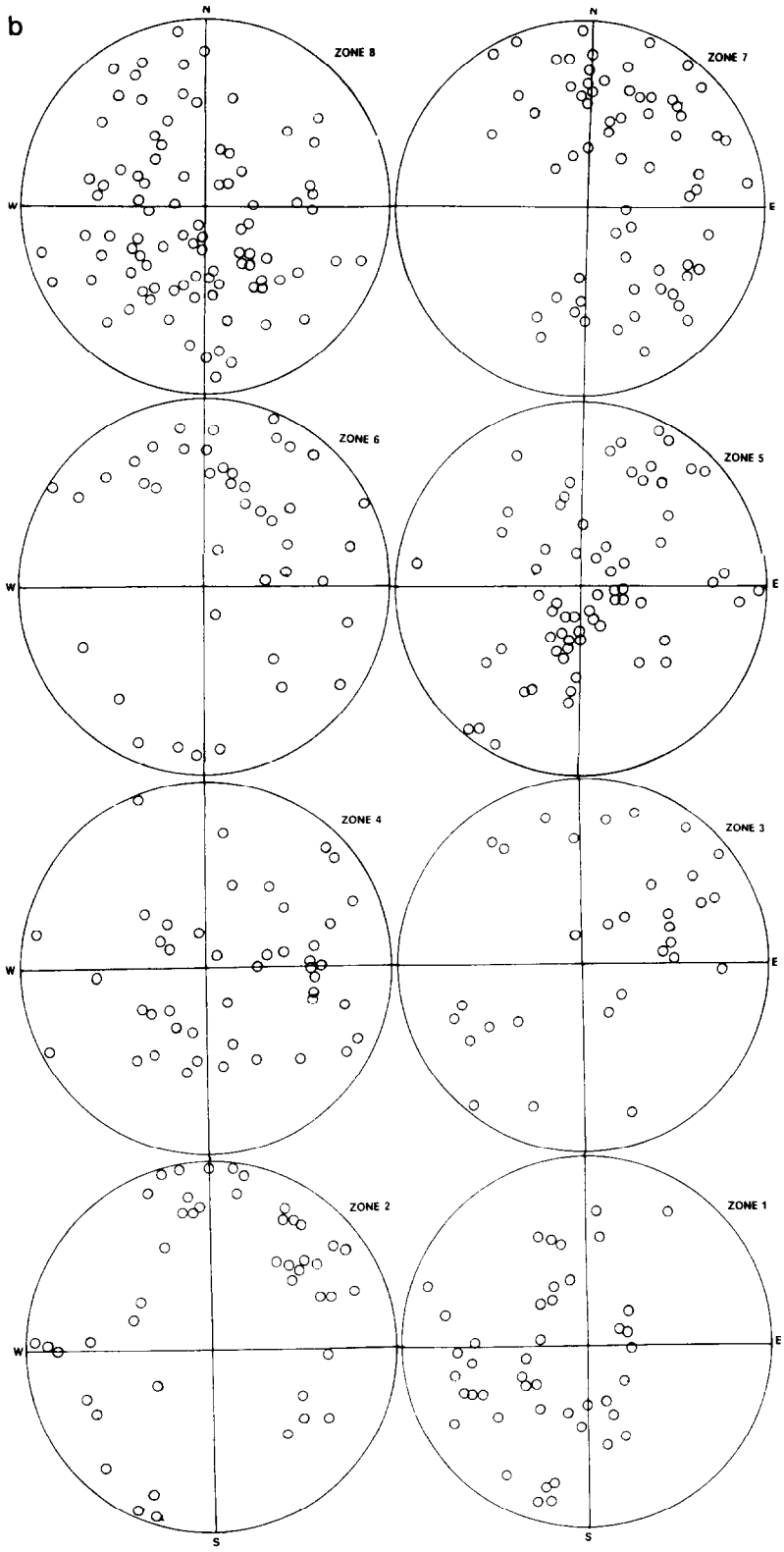


Fig. 14 (continued).

shear (Westbrook, 1974; Coward and Park, 1987). In general, the magnitude of anisotropy is lower in the central zones 4 and 5. Dearnley (1962) considered that this area preserves relict Scourian structures and has escaped major Laxfordian reworking. Although no such simple solution has subsequently emerged (Coward, 1973; Graham and Coward, 1973), there is a general indication here that these zones have escaped the major fabric-imparting events which affected zones to the north and south. Susceptibility-ellipsoids are more often oblate ($F > L$ foliated) than prolate ($L < F$ lineated).

The predominance of foliated anisotropies is reflected in weak girdle distributions of directions of maximum susceptibility in the northern zones (Fig. 14a). NW–SE trends (notable in zone 7) reflect the plane of maximum Laxfordian strain. Exceptions are zones 3, 4 and 5 where significant numbers of maximum and/or minimum susceptibility directions swing around to N and NE directions compatible with planes of earlier Scourian strain. Directions of minimum anisotropy (Fig. 14b) exhibit few apparent trends, except that below zone 7 they tend to avoid the NW–SE quadrants containing the plane of maximum Laxfordian strain.

As with comparable studies of metamorphic rocks (e.g. Hrouda and Janak, 1976; Hrouda et al., 1988), the Hebridean rocks tend to possess appreciable anisotropies of variable magnitude and direction. The measured effect is a bulk one recording the total contribution of the paramagnetic silicates as well as the ferromagnetic phases. It is anisotropy of *susceptibility* and not anisotropy of *remanence*.

The contribution of the small amounts of magnetite responsible for the stable remanence is unclear. Only a few percent or less of the total magnetite may carry the stable remanence and this is likely to represent a proportion of the blade-like fracture fillings (Hay et al., 1988). Intergranular magnetites, at least, will have shapes constrained by alignment of pre-existing silicates and some deflection of the remanence vectors by the effects of anisotropy is to be anticipated. This problem is considered to be less severe in medium- and high-grade rocks even when they show quite strong anisotropies of susceptibility (Irving and Park, 1972). Variable orientations of the anisotropy ellipsoids (Fig. 14) imply that anisotropy contributes to overall scatter of remanence, but will show only small systematic effects constrained predominantly by the Laxfordian strain.

Palaeomagnetic poles

Prior to Laxfordian times the exposed terrane in the Outer Hebrides was at temperatures above the magnetite Curie point (Dickinson and Watson, 1976; O'Hara, 1977). Metamorphic assemblages indicate that temperatures in the range 700–500 °C during this episode dropped rapidly to 300–200 °C by ca. 1600 Ma (Dickinson and Watson, 1976) as the crustal section was elevated from depths of the order of 10–20 km or less. Hence if remanence is related to this initial episode of crustal uplift, correlation should be sought with APW between 1800 and 1600 Ma. Later episodes of basement reheating were due to burial beneath (1) the Torridonian sedimentary pile which attained a thickness of up to 7 km at ca. 1100–1000 Ma (Smith et al., 1983) and (2) the lateral extension of the Caledonian nappe pile in Ordovician times. A further possible cause of later overprinting is Mesozoic rifting which created extensional basins on both sides of the Outer Hebrides. The maximum temperature attained during these younger events was probably a thermal high of ca. 275 °C beneath the Moine Thrust during Caledonian times. Even if maintained for a period of several tens of millions of years, such temperatures would be incapable of unblocking Lewisian remanence components isolated above 500 °C by the thermal demagnetisation (Pullaiah et al., 1975). Furthermore, the magnetic imprint in the Mainland Lewisian is essentially the same as that in the Outer Hebrides (Fig. 10) although the latter could only have been buried beneath a thinner lateral continuation of the Caledonian nappe pile. The theory of magnetic unblocking assumes that it is a purely physical process. Its conclusions are not valid if chemical processes also play a part. However, since palaeomagnetic conglomerate tests on

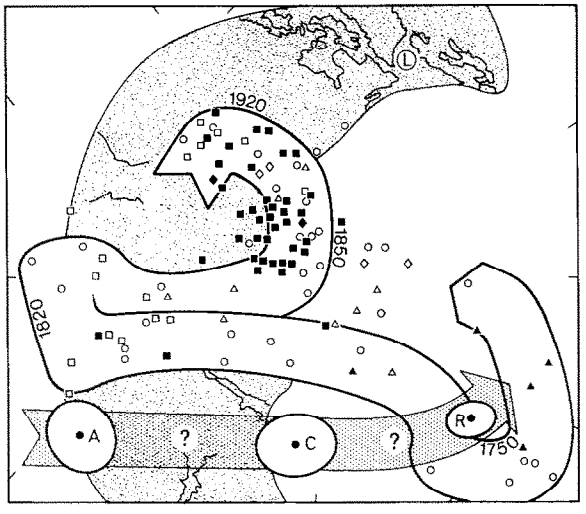


Fig. 15. APW path for the continental crust between ca. 1900 and 1700 Ma after Piper (1987b). Lewisian poles from inferred relict remanences are *A* ('A1' components) and *C* ('C' components) and *R* is the pole derived from the shallow N–S axis identified on Raasay (Tables 3 and 5). Lewisian poles are adjusted to their inferred Proterozoic configuration relative to the Laurentian Shield by closing up the North Atlantic to correct for Phanerozoic drift. For a key to the other poles shown in this figure see Piper (1987b).

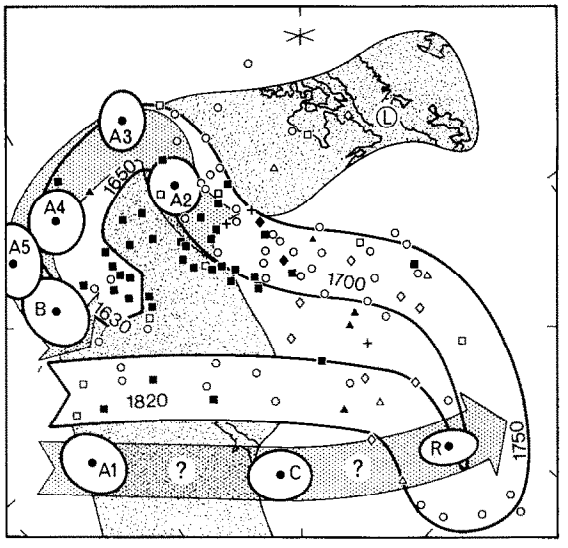


Fig. 16. APW path for the continental crust between ca. 1700 and 1630 Ma after Piper (1987b) with the poles derived from the NW directed magnetisations in the Lewisian terrane ('A2' to 'B' magnetisations). Pole positions are adjusted to their inferred Proterozoic reconstruction relative to the Laurentian Shield using rotational operations given in Piper (1987b).

Lewisian clasts within basal Torridonian sediments show random magnetisations (Piper and Poppleton, 1990) and the Mainland and Hebridean magnetisation records are comparable, it is concluded that the Outer Hebrides also preserves a pre-Torridonian remanence.

Prior to opening of the North Atlantic Ocean Basin in late Mesozoic and Tertiary times, NW

TABLE 4

Summary of magnetic component properties within the Lewisian metamorphic terrane, Outer Hebrides

	No. of sites/ samples	1 % defining components	2 Mean coherency parameter	3 Mean site α_{95}	4 Mean definition parameter
Zone 8 (NE)	34/268	82.0	55.5	18.1	6.4
7	17/157	73.8	54.3	18.8	5.8
6	15/129	62.7	56.7	18.4	6.2
5	21/163	82.8	68.1	14.8	6.1
4	16/131	73.3	35.4	17.5	5.9
3	16/123	85.4	91.4	17.7	6.9
2	15/112	87.5	83.7	16.1	7.9
1 (SW)	14/138	97.1	90.6	12.8	8.7
Raasay	20/192	75.0	75.7	14.6	7.3

Column 1 is the percentage of samples in each collection defining components with $n \geqslant 3$ and $\alpha_{95} < 25°$. Column 2 is the percentage of the collection defining site mean directions (again with $n \geqslant 3$ and $\alpha_{95} < 25°$). Column 3 is the mean α_{95} value of these sites and column 4 is the mean of the number of demagnetisation steps defining the sample remanence components within each zone.

Scotland formed a peripheral segment of the Laurentian Shield (Bridgwater et al., 1973). The palaeomagnetic record for the interval 1800–1600 Ma from this shield is based largely on data from the "Hudsonian" metamorphic terranes broadly contemporaneous with the Laxfordian. They record widespread tectono-thermal reworking of the crust during these times and include the Churchill, Circum-Ungava and other terranes peripheral to the Superior Craton of the Canadian Shield, and the Nagssugtoqidian Belt of central Greenland. In the Fennoscandian Shield, this interval is represented by the Svecokarelian and Svecofennian episodes which were widely accompanied by syn- and post-tectonic magmatism. Palaeomagnetic poles from these terranes cross the Laurentian Shield firstly from west to east at ca. 1850–1750 Ma and then from east to west at ca. 1750–1650 Ma (Figs. 15 and 16). Much of the latter path is recorded as a continuous sequence of uplift magnetisations within the Nagssugtoqidian terrane of west Greenland (Piper, 1981, 1985). Although the polar record is fairly detailed (Fig. 16) the bulk of the magnetisations are derived from slowly cooled plutonic terranes and are correspondingly difficult to date. By considering the evidence from Greenland, Piper (1985) considers that the best estimate for the age of the SE → NW swathe crossing the Laurentian Shield is 1750–1650 Ma. It is then immediately preceded by the palaeomagnetic record from the Coronation geosyncline and Authapuscow aulacogen bordering the Slave craton (Irving and McGlynn, 1979) and part of the Aphebian record from the Circum-Ungava geosyncline (Schmidt, 1980). For further details of this APW track and a key to the poles plotted in Figs. 15 and 16, see Piper (1987).

Palaeomagnetic poles derived from the 'A2' to 'A5' and 'B' magnetisations in the Lewisian terrane of the Outer Hebrides (Tables 2 and 5) commence near the younger extremity of the path defined by the Nagssugtoqidian data when Greenland and NW Scotland are adjusted for Phanerozoic drift (Bullard et al., 1965). They define a loop which terminates at a group of poles including younger magnetisations dated ca. 1630 Ma from the Nagssugtoqidian Belt and elsewhere. This correlation dates the predominant continuum of magnetisations within the Lewisian Complex at ca. 1700–1630 Ma, although a somewhat younger age of ca. 1550 Ma would not be incompatible with the position of the central part of this track (Piper, 1987b). It suggests a genetic link with the widespread retrogressive amphibolite facies metamorphism in the Laxfordian terranes (Park and Tarney, 1987; Hay et al., 1988), and would be compatible with remanence acquisition during the period of rapid uplift accompanying this metamorphism (Dickinson and Watson, 1976). The record is continuous with, but does not precisely correspond to, the palaeomagnetic imprint in the Nagssugtoqidian, a belt which continues into the Lewisian terrane in the pre-drift configuration (Bridgwater et al., 1973). Since these are both plutonic terranes which may well be exposed to differing crustal levels, no exact correspondence between the palaeomagnetic records from the two terranes is to be expected.

The precursor path to the 1750–1630 Ma swathe in Fig. 16 is less well understood. The evidence is discussed in Piper (1987b) and will not be reiterated here. Poles from the 'A1' and 'C' remanences and the component unique to Raasay plot in sequence close to the path between ca. 1820 and 1750 Ma (Fig. 15). The deflection of the remanence by the magnetic anisotropy noted in the previous section might explain the failure of these poles to correspond precisely to the predicted path. These directions lie at a high angle to the NW–SE plane containing the maximum and intermediate directions of Laxfordian principal strain. The 'C' component is most prominent in the north (Fig. 17) where this strain is most evident. Alternatively 'A1' and 'C' may have been subjected to late Laxfordian deformation. 'A2' and 'A5' and 'B' components lie closer to the plane of Laxfordian strain and the influence of this effect will be less evident.

This interpretation of the 'A1' and 'C' components implies acquisition near to the peak of the Laxfordian metamorphism. Two regional observations suggest that 'A1' and 'C' components are relict features which have survived the later magnetising episode. 'A1' is preserved in a kernel of Scourian rocks in the Southern Zone of the Mainland where it is replaced firstly at reworked

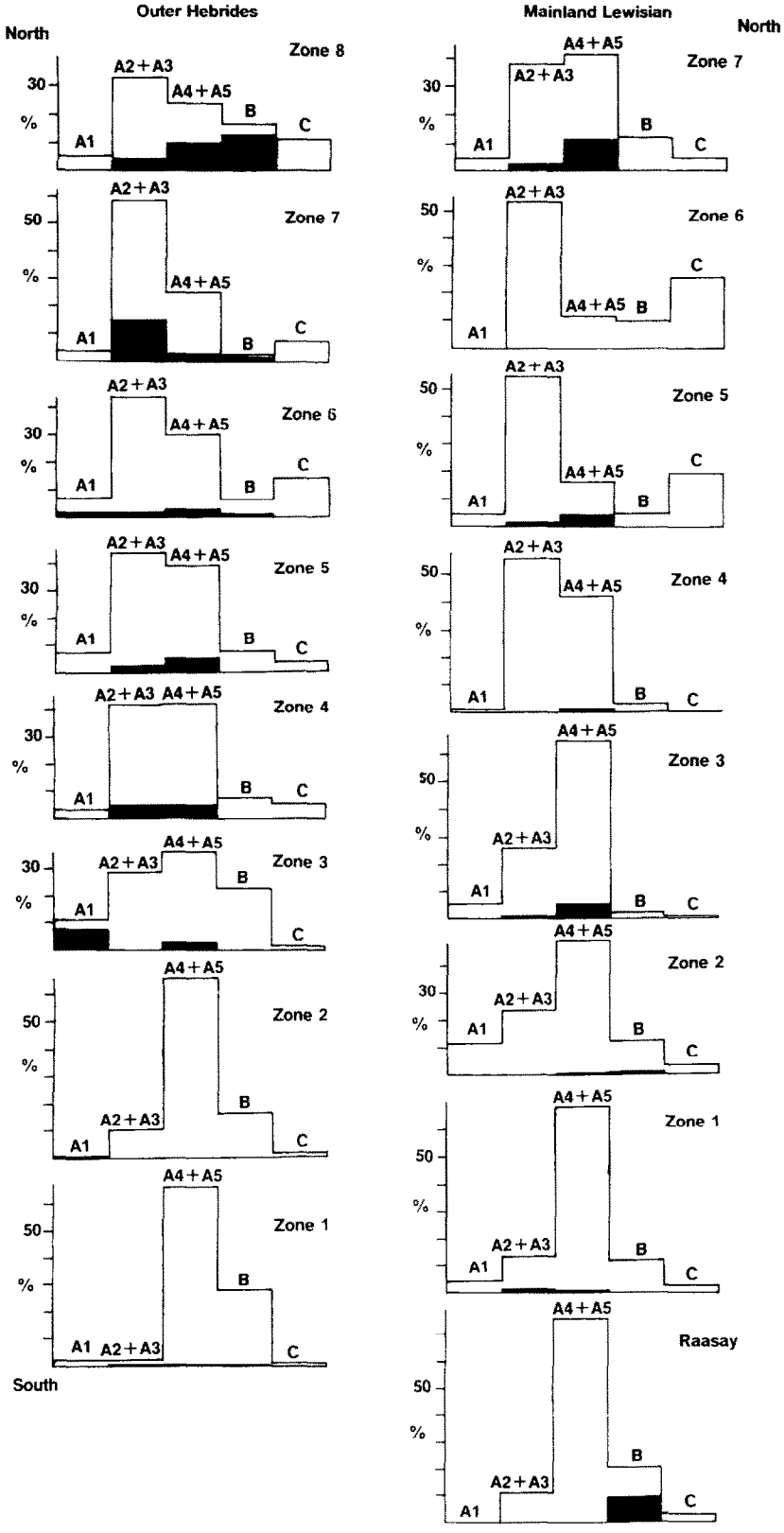


Fig. 17. The proportions of magnetisations within each of the eight subdivisions of the Mainland and Hebridean outcrops (Fig. 1) falling into each of five divisions ('A1', 'A2' and 'A3', 'A4' and 'A5', 'B' and 'C'). The black sectors are reversed equivalents of each component.

TABLE 5

Palaeomagnetic directions and pole positions from the Lewisian terrane, Outer Hebrides and Raasay; tilt adjusted and relative to the Laurentian Shield

	D	*I*	*D* *	*I* *	°E	°N	°E +	°N +
A1	95.1	68.4	98.7	64.7	47.9	34.0	10.2	33.2
A2	292.9	47.8	292.3	51.8	264.1	38.6	226.1	39.6
A3	325.0	38.1	326.5	41.7	222.7	48.7	183.9	49.5
A4	315.3	−3.0	315.3	0.9	222.6	22.7	184.3	23.5
A5	316.6	−28.4	316.0	−24.6	216.5	10.7	178.3	11.4
B	298.4	−29.6	298.5	−25.6	231.8	2.9	193.7	3.8
C	32.4	40.3	35.8	40.0	121.2	46.6	82.5	45.8
Raasay	359.0	2.1	–	–	175.2	33.4	136.5	33.5

* Tilt adjusted; + relative to the Laurentian Shield.

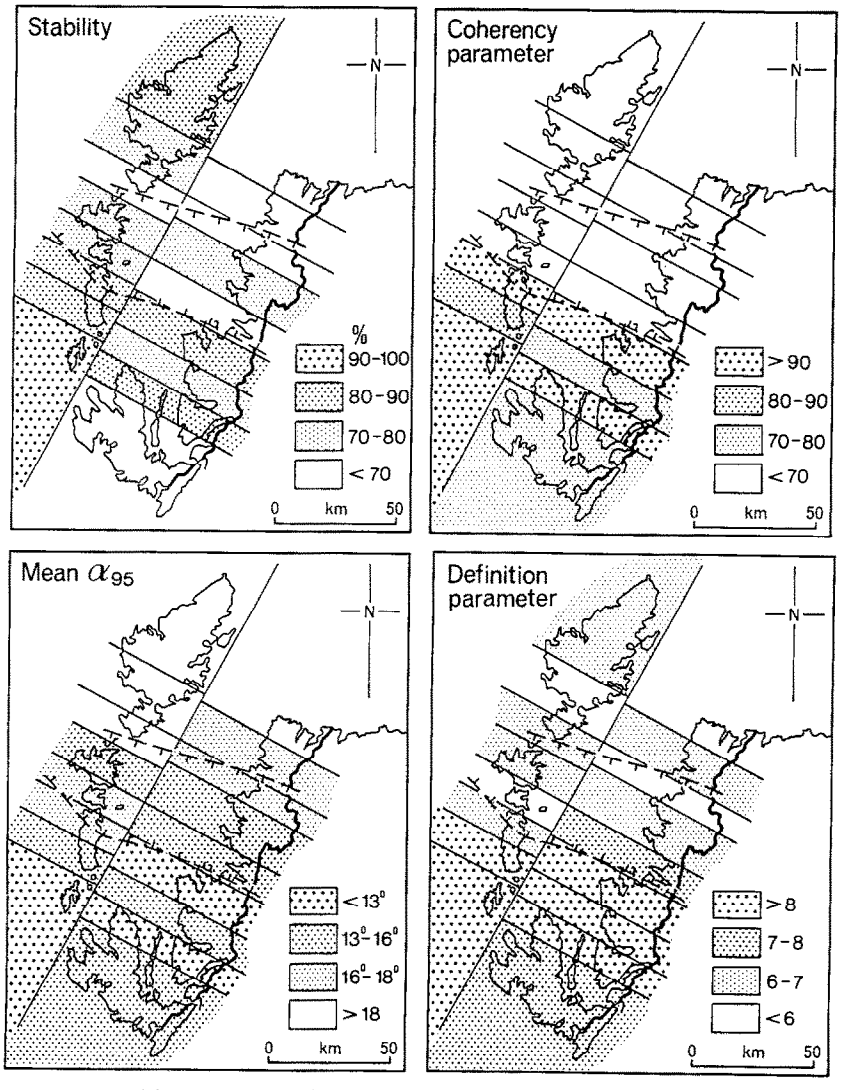


Fig. 18. Four parameters summarising properties of magnetic remanence in the Lewisian outcrop corrected for 90–100 km of presumed Caledonian sinistral motion in the Minches. The definition of each parameter is given in the text and the caption to Table 4. The tagged line embraces the Central Zone of relict Scourian structures on the Mainland and the continuation into the Hebridean outcrop proposed by Dearnley (1962).

Scourie dyke margins, and then throughout the dykes, as zones of Laxfordian reworking are entered (Poppleton and Piper, 1989). 'C' components are confined largely to the Central Zone on the Mainland (Fig. 17) where pre-Laxfordian structures are preserved unreworked. They are largely absent in Laxfordian terranes to the north and south. However, this interpretation is not unambiguous. The 'A1' and 'C' components also reside in the lowest blocking temperature and lowest coercivity components. This presumably reflects the state of the magnetite grains and implies that they would also be most sensitive to later overprinting events. Hence it is possible that these remanences are preserved only where the magnetite has proved susceptible to remagnetisation. Failure to identify these components in Torridonian conglomerates (Piper and Poppleton, 1990) tends to argue against this possibility.

Regional variations in magnetic imprint

The manner in which the Lewisian remanence changes from NE to SW across the outcrop is shown in Fig. 17 in terms of the relative proportions of sample components falling into five divisions. Small proportions of 'A1' and 'B' components occur throughout both sectors and exhibit no obvious regional variations. 'C' components are most abundant in the northern three zones (6, 7 and 8) in the Hebrides and in northern zones (5, 6 and 7; zone 8 has too few data points for analysis) on the Mainland. The NW + ('A2' and 'A3') and NW shallow to negative ('A4' + 'A5') components show a striking and reciprocal variation along each outcrop: 'A2' + 'A3' are most abundant in the north and replaced southwards by 'A4' and 'A5'. 'A4' and 'A5' become more abundant than 'A2' and 'A3' at zone 4 in the Hebrides, and between zones 3 and 4 on the Mainland. Thus the variations in the Hebrides mirror those on the Mainland but are offset in a sinistral sense with respect to them. This identifies a strike-slip fault in the Minches Basin with a sinistral motion of approximately 95 km.

When adjustment is made for this motion, general correlation is achieved between magnetic properties on either side of the fault zone (Fig. 18 and Table 4). Greater percentages of samples tend to be stable (components defined by three or more demagnetisation steps) in the south of each outcrop than in the north. This reflects increased percentages of the stable, higher blocking-temperature (especially 'A4') components in the south (Fig. 18a), and hence the greater intensity of this later magnetising event here.

A parameter of relevance to palaeomagnetic study of metamorphic terranes is the within-site coherency, the percentage of samples within a site defining a single predominant direction. It attains its highest value in the southern zones of each outcrop (Fig. 18b) and also reflects the intensity of this later overprinting event. Further north directions, although stable, tend to be more scattered and are less likely to show within-site groupings.

The quality of definition of within-site directions, as expressed by the mean α_{95} value, also tends to be higher in the south of each outcrop (Fig. 18c). The definition parameter, expressed as the average number of demagnetisation steps yielding the magnetisations, also tends to be higher and reflects the higher stability of components here. There is a general correspondence when adjustment is made for sinistral strike-slip movement in the Minches.

Figures 17 and 18 show that the palaeomagnetic imprint in the Hebridean and Mainland basement predates this sinistral motion. However, it postdates the late tectonic (Laxfordian) episode because it shows no direct correlation with the zones of relict Scourian or Laxfordian reworking, either on the Mainland or in their postulated continuation into the Outer Hebrides (Fig. 18; Dearnley, 1962).

Phanerozoic dyke swarms

Dolerite dykes cut the Precambrian basement in the Outer Hebrides most prominently in the central part of the archipelago. This is on the strike continuation of the Skye igneous centre and they are interpreted (Jehu and Craig, 1923, 1927) as a lateral extension of the British Tertiary igneous province of early Tertiary age (ca. 55 Ma). They are included in contact tests (e.g. Fig.

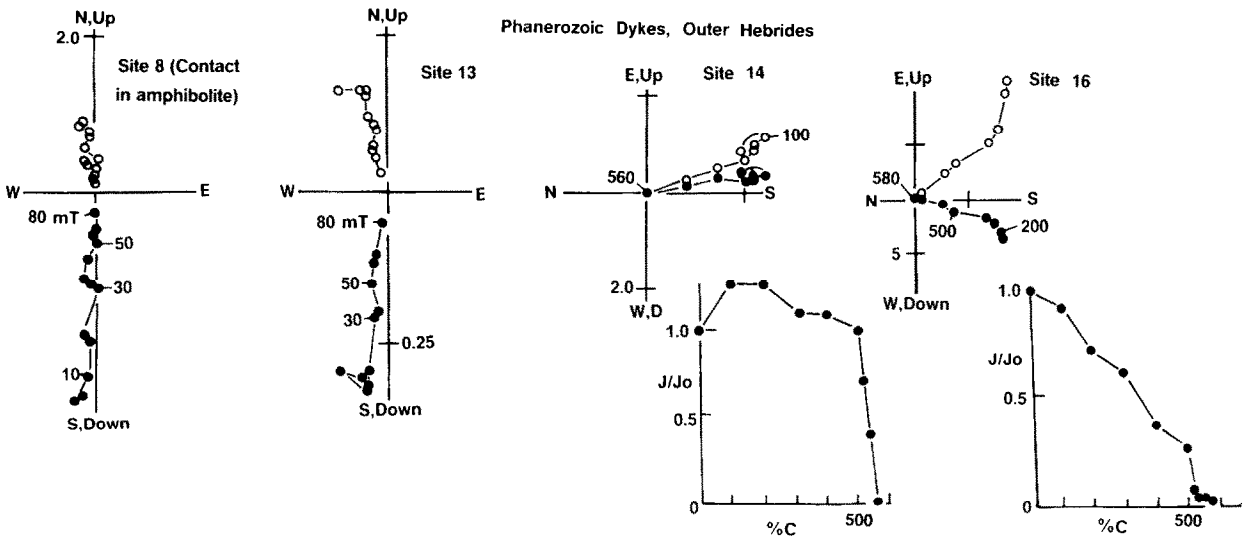


Fig. 19. Examples of a.f. and thermal demagnetisation of Phanerozoic dykes cutting Lewisian basement on Barra (zone 8 of Fig. 1) and from a contact with country-rock amphibolite. Symbols are as for Fig. 6.

8) and comprise sites 49 and 54 in South Uist and 21 on Barra; the common reversed polarity (Table 1) is similar to the majority of dykes on Skye.

Pre-Tertiary dykes are found as far north as South Harris (Jehu and Craig, 1927). They trend NW–SW to E–W and comprise either ophitic quartz dolerites exhibiting marginal alteration of pyroxene to amphibole, or lamprophyric products (mostly camptonites) of a more hydrous magma (Jehu and Craig, 1923). In places they are cut by members of the Tertiary swarm. On petrographic grounds they are linked with either the Old Red Sandstone (Devonian) minette, kersantite and spessartite lamprophyres of the NW Highlands of Scotland or with the (? Permian) camptonite–monchiquite lamprophyres of the SW Highlands (Dearnley, 1962). They have been sampled at nine locations on Barra (Fig. 4).

Southerly shallow negative directions residing predominantly, or entirely, in one component are resolved by a.f. and thermal demagnetisation (Fig. 19). These directions (section IX in Table 1) contrast with the NW components in the host gneisses and amphibolites (section VIII, Table 1) except at site 1 where Palaeozoic remagnetisation is possible. Site mean directions are compared in Fig. 20 with the reversed palaeofield migration path for this area computed from the APW path for Eurasia since 310 Ma (Piper, 1987b). Site mean directions on Barra have steeper negative inclinations than components resolved from quartz dolerite dykes in the North Highlands (Esang and Piper, 1984) and dated 320 Ma (mid-Carboniferous) in age. The group mean remanence direction falls on the palaeofield migration path at 260 Ma compatible with a mid-Permian age (Fig. 20) and consistent with the camptonite

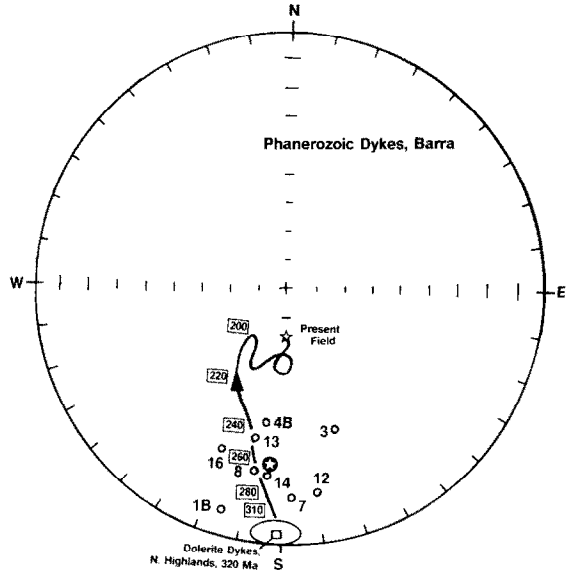


Fig. 20. Site mean directions of magnetisation from Phanerozoic dykes on Barra (section IX, Table 1) compared with the palaeofield migration path for this area calculated from the Phanerozoic path of Eurasia (after Piper, 1987b).

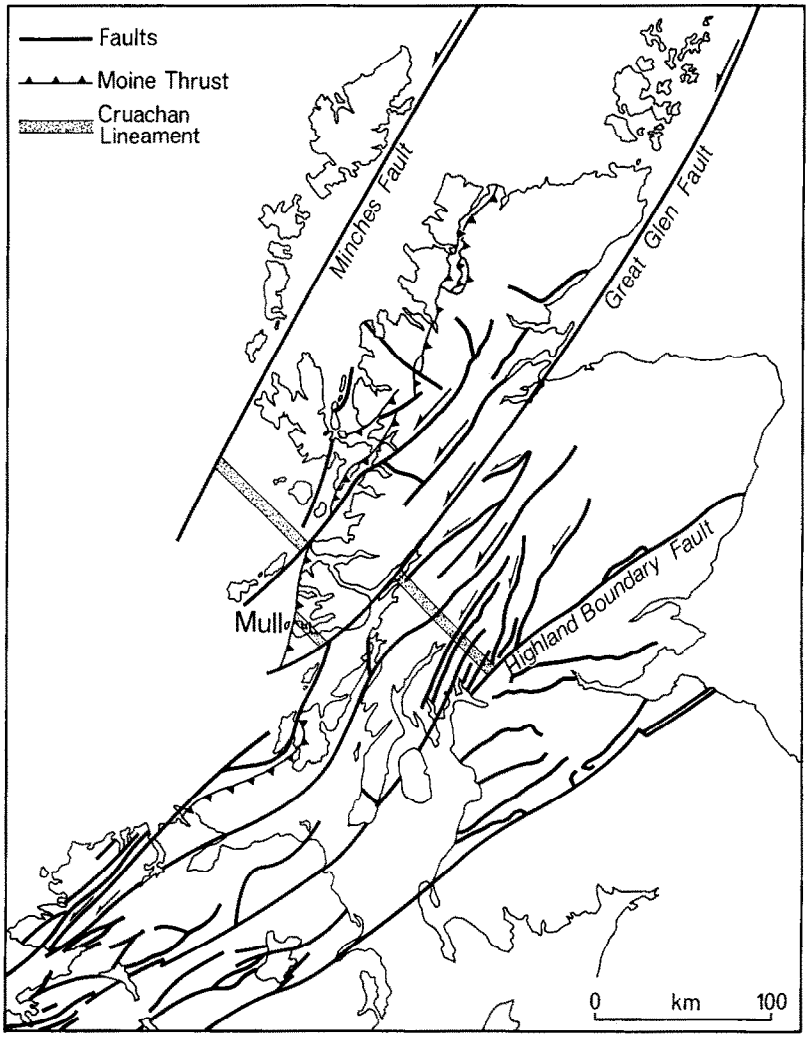


Fig. 21. The Lewisian terrane and adjoining part of the Caledonian orogenic belt in Northern Britain showing major Caledonian strike slip dislocations after Pitcher (1969) and Hutton (1982). The Minches Fault adjusted for the sinistral motion identified in this study is shown for comparison.

mineralogy of younger intrusive suites in the Highlands. The uniform reversed polarity is compatible with magnetisation during the Permo–Carboniferous superchron. The dykes are a late expression of a regime dominated by N–S directed tensile stresses (Anderson, 1955) before extensional tectonism developed in the Minches Basin between the Hebrides and Mainland in Permo–Triassic times along NE–SW directed faults (Brewer and Smythe, 1986).

The Minches Fault

The fault displacement identified from displacement of the magnetic imprint in the basement defines a sinistral strike-slip fault zone sited within the Minches Basin and trending approximately N35° E. This trend links the fault zone to the system of linear NE–SW trending faults sited within the Scottish Caledonides and extending SW into Ireland (Pitcher, 1969; Fig. 21). The Minches Fault is confined to the Precambrian basement. Parallel zones of cataclasis are found within Lewisian rocks on the islands of Coll and Tiree to the SE (Dearnley, 1962), but the remaining examples have all been described from the Orthotectonic Caledonides (Anderson, 1955). Most have sinistral motions of up to a few kilometres with downthrows to the SE, although dextral motions are also observed on some members (Pitcher, 1969).

The outstanding member of this suite (and the

only one with a strike continuation throughout the Caledonides) is the Great Glen Fault. Many models for the movement history on this fault have been proposed since Kennedy's classic paper (1946) and none are entirely satisfactory. The best constraint appears to be the offset of a basement structure identified by geophysical evidence and known as the Cruachan lineament (Hipkin and Hussain, 1983). It is interpreted as a deep crustal vertical shear zone concealed beneath Caledonian nappes. The offset is equivalent to a sinistral strike-slip of about 25 km in the Lewisian basement (Hall, 1986). The basement is overthrust both to the north and the south of the fault zone by the Caledonian orogen and Hall (1986) estimates that a comparable motion was taken up by the Moine Thrust. This yields a resultant motion of the Northern Highlands of about 50 km with respect to the Grampian Highlands to the SE of the fault zone. The sharp 20° bend in the trace of the Great Glen Fault towards a more ENE–SWS course south of Mull (Fig. 21) also favours a more limited interpretation of the movement history on this fault plane (Hall, 1986).

Geological evidence suggests that major displacements on the NE–SW fault system are a late Caledonian phenomenon and post-date movement on the Moine Thrust Zone at ca. 430 Ma (Watson, 1984). Within the Caledonides, they have a close spatial relationship to the "Newer" granite suite. Some granites were emplaced along existing faults, while others are fractured and offset by them (Watson, 1984). This is most graphically illustrated in Ireland where emplacement of the Donegal granite at 400 Ma was synchronous with sinistral motion (Hutton, 1982). The Lennan Fault in NW Donegal, a possible continuation of the Great Glen Fault (Fig. 21), has a sinistral displacement of at least 40 km and motion was in progress during deposition of Lower Devonian sediments.

More limited motions occurred in Middle Devonian, Carboniferous (Hercynian) and later times (Pitcher, 1969). A similar history is therefore likely to have characterised the Minches Fault. Recognition by this study of a mid-Permian dyke suite on Barra in direct continuity with the camptonite–monchiquite swarm of the Morven Centre on the Mainland (Dearnley, 1962) is evidence that major movements had ceased by Permian times.

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